

# Information Elicitation by Ordinal Agreement

Yiling Chen<sup>1\*</sup>, Shi Feng<sup>1</sup>, Grant Schoenebeck<sup>2</sup>, and Fang-Yi Yu<sup>3</sup>

<sup>1</sup> Harvard John A. Paulson School of Engineering and Applied Sciences, Harvard University, Cambridge, MA, USA

<sup>2</sup> School of Information, University of Michigan, Ann Arbor, MI, USA

<sup>3</sup> Department of Computer Science, George Mason University, Fairfax, VA, USA

**Abstract.** Ordinal information is central to tasks ranging from conference peer review to Reinforcement Learning from Human Feedback for large language models. In many such settings, there is no objective ground truth against which reports can be directly verified, making incentive-compatible elicitation challenging. Peer-prediction mechanisms have been designed to achieve truthful elicitation despite the lack of ground-truth verification. However, existing peer-prediction methods do not provide incentives for agents to truthfully compress rich private information (e.g. reviews) into coarser ordinal reports (e.g. rankings) requested by the mechanisms. We propose a simple ordinal peer-prediction mechanism that rewards agents based on pairwise agreement, measured by Kendall tau. Under a broad class of Bayesian Stochastically Order Consistent models—which includes classical models such as Mallows and Thurstone—we prove that truthful reporting is stochastically dominant, and strongly truthful under index-equivariant strategies. This provides robust incentives even for agents with non-linear utility functions, such as risk aversion. Our mechanism further accommodates heterogeneous agents, top-K reports, and subset assignments. Experiments on synthetic Mallows and Thurstone data, together with a real-data stress test on the SUSHI preference dataset, show right-shifted truthful-payment distributions and the highest normalized sensitivity among tested baselines.

## 1 Introduction

Many important evaluation tasks require eliciting ordinal information. Conference program committees rank submissions for acceptance; platforms collect human feedback to compare the quality of LLM responses; and instructors use peer assessment to ask students to rank a small set of assignment submissions. These settings share a common structure: (1) Evaluators often have informative but imperfect signals to judge relative quality. (2) Investing more effort leads to higher-quality judgments. (3) There is often no objective ground truth against which reports can be directly verified.

This absence of ground truth creates a fundamental incentive problem. Without appropriate incentives, agents may invest little effort in evaluation or report strategically. Directly rewarding “correct” answers is not possible when correctness cannot be verified. The peer-prediction literature addresses the challenge of elicitation without ground truth by rewarding agents based on the statistical relationships among their reports. When properly designed, such mechanisms make truthful reporting an equilibrium even in the absence of verifiable outcomes. However, most existing peer-prediction mechanisms require agents to report all of their private information [22,15], typically as a numerical or categorical signal, and their incentive guarantees can fail when agents observe richer information than they are asked to report [8]. This creates a mismatch in many evaluation settings: agents may observe rich private information, such as detailed reviews, latent quality assessments, or noisy cardinal scores, while the designer only wants a coarser ordinal report, such as a ranking or a set of pairwise comparisons.

This paper studies peer prediction for coarse ordinal elicitation. Agents may observe rich private information about item quality, but the mechanism asks them to report only ordinal information. We allow several natural ordinal report formats, including full rankings, fixed-shape bucket orders such as top- $K$  sets and tiered reports, and rankings over assigned subsets of items. Our goal is

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\* Authors are listed in alphabetical order.

to design incentives for truthful ordinal reporting without requiring agents to reveal their full private information. To do so, we leverage the fact that ordinal reports naturally encode pairwise comparisons among items.

This suggests a simple agreement-based mechanism: reward agents according to whether their reported pairwise comparisons agree with those of their peers, equivalently through a Kendall tau-based pairwise-agreement score. Such an approach is attractive because it is simple, transparent, and directly aligned with the ordinal information the designer wants to elicit. In general elicitation settings, however, rewarding agents for agreement can encourage them to suppress private information when they believe their signal is likely to be in the minority [14]. For example, if an evaluator believes that most peers will mark nearly every submission as acceptable, then a reward based only on agreement may induce the evaluator to report “acceptable” even for a submission they privately judge to be weak. This is why agreement mechanisms typically require strong self-dominance conditions — after observing a signal, an agent must believe that other agents are more likely to report the same signal than any alternative [32] — to have a truthful reporting equilibrium.

Our key observation is that, although such self-dominance conditions are generally strong for arbitrary signals, they become natural at the level of pairwise ordinal comparisons. We identify a broad class of Bayesian Stochastically Order Consistent information structures under which each truthful pairwise comparison is more likely to agree with a peer’s comparison than to disagree, even after conditioning on the agent’s richer private information. Under this condition, the Kendall tau mechanism makes truthful ordinal reporting stochastically dominant, despite eliciting only a coarse ordinal summary of the agent’s information.

*Contributions* We introduce a peer-prediction mechanism for coarse ordinal reports based on Kendall tau distance and its pairwise-agreement extension. The mechanism applies to several natural reporting formats, including full rankings, fixed-shape bucket orders [5], and rankings over assigned subsets of items. The bucket-order setting covers full rankings and top- $K$  sets as special cases and also captures tiered reports. Its reward rule is simple and transparent: agents are paid according to agreement with peers on the pairwise comparisons induced by their reports. This simplicity is useful both theoretically and practically. When evaluators are already comparing a small set of alternatives, ordinal reports can be easier to specify and audit than calibrated cardinal scores, consistent with a long line of work in psychometrics on comparative judgment [28,27,20].

We provide strong incentive guarantees for this mechanism under a class of Bayesian Stochastically Order Consistent (Bayesian SOC) information structures. Bayesian SOC ensures that each truthful pairwise comparison is more likely to agree with a peer’s comparison than to disagree, even after conditioning on the agent’s richer private information. This class includes classical ranking information structures, such as the Thurstone and Mallows models. Under Bayesian SOC, truthful ordinal reporting is stochastically dominant: truth-telling not only maximizes expected payment, but yields a payment distribution that first-order stochastically dominates the distribution under any unilateral deviation. This is stronger than standard equilibrium guarantees based only on expected utility and provides robustness to agents with nonlinear utility functions, such as risk aversion. Moreover, we show that the mechanism is strongly truthful if agents’ strategies cannot coordinate on an external indexing of items.

We also study the measurement accuracy of the mechanism, namely how well payments track the quality of an agent’s report [3]. This property is important in practice for fairness [2], motivation [30], and trust in the mechanism. Prior work has shown that strong stochastic-dominance incentives can come at a substantial cost to accuracy [32]. In contrast, our experiments on synthetic data from Mallows and Thurstone models, together with a real-data stress test on the SUSI preference

dataset [11], suggest that the Kendall tau mechanism remains sensitive to report quality in the ordinal settings we study.

## 1.1 Relation to prior work.

Peer prediction began with mechanisms such as Bayesian Truth Serum [22] and has developed into a broad literature on eliciting unverifiable information from correlated reports. A central difficulty is that simple agreement can make agents coordinate on common or focal reports rather than reveal their private information [14]. Many later mechanisms therefore reward reports through prediction, correlation, or mutual-information-style quantities, including informed-truthful and multi-task mechanisms [25, 16, 13, 23, 7]. These mechanisms typically operate on categorical or numerical reports, often require agents to report a sufficiently rich representation of their signal, or learn/estimate statistical relationships across tasks. Our mechanism instead asks for ordinal reports and uses a fixed, transparent pairwise-agreement score.

Our setting connects most directly to the bonus-penalty payment (BPP) mechanism for binary pairwise comparisons [4], which operates under Bayesian Strong Stochastic Transitivity (Bayesian SST). BPP has a three-agent structure: an agent evaluates one pair and is scored using signals from two other agents on related pairs. This makes BPP well suited to sparse pairwise-comparison assignments. By contrast, our mechanism asks agents for richer ordinal reports—full rankings, top- $K$  sets, bucket orders, and subset assignments—and scores them by all-pairs agreement on overlapping item pairs. Thus our framework is less general in assignment structure, because pairs must be reviewed by more than one agent, but it handles richer reports directly.

The modeling assumptions are also different. BPP is designed for homogeneous pairwise signals, whereas Bayesian SOC allows heterogeneous agents and full signal vectors; for instance, different reviewers may have different noise levels in a Thurstone or Mallows model. When SOC and SST are viewed as conditions on the same full-signal distribution, SOC is stronger because it imposes pointwise swap inequalities on full signal vectors, while SST is a pairwise condition on comparison probabilities. However, the Bayesian SOC framework and the BPP/SST framework are not directly nested as modeling frameworks; we give a more careful comparison in Appendix F.

Our stochastic-dominance guarantee is related to recent stochastically dominant peer prediction for multi-task categorical reports [32]. Their mechanisms use task partitions to obtain stochastic dominance for categorical signals. Here, the ordered structure of rankings and Bayesian SOC allow a simpler Kendall tau-based pairwise-agreement score: the proof turns a deviation into adjacent pair corrections, each improving the score distribution. Technically, the proof controls weighted peer-alignment margins rather than independent categorical tasks.

Conceptually, our framework also connects to property elicitation [17]: the target function maps a rich private signal vector to the ordinal property we want to elicit, such as a ranking, top set, or bucket order. From this perspective, Kendall tau-based pairwise agreement is a scoring rule for an ordinal projection of the agent’s information rather than for the full signal. This distinction is important in settings such as review, peer assessment, and RLHF, where the platform often wants a tractable ordinal judgment even though agents may observe richer cardinal evidence.

*Paper organization.* Section 2.1 introduces the Bayesian Stochastically Order Consistent (SOC) framework and Section 2.2 formalizes ordinal peer prediction. Section 3 proves the full-ranking Kendall tau mechanism is truthful and stochastically dominant truthful. Section 4 gives the subset-assignment and bucket-order extensions, the latter also implying the top- $K$  result. Section 5 reports the main empirical comparison; proofs, specialized top-set details, and additional experiments are in the appendices.

## 2 Problem Formulation

We consider a collection of  $M$  items and a set of  $N$  strategic agents. Each agent privately observes noisy signals for the items. Our goal is to design mechanisms to elicit agents' private ordinal information truthfully. We first introduce the joint distributions of agents' private signals in Section 2.1 and then define the information elicitation problem in Section 2.2.

We first define Kendall tau distance between two item rankings. Given two items  $k$  and  $l$  and an ordering  $r$  of them, the sign function is

$$\text{sign}(r(k), r(l)) = \begin{cases} 1 & \text{if } r(k) < r(l) \\ -1 & \text{if } r(k) > r(l) , \\ 0 & \text{otherwise} \end{cases}$$

where  $r(k)$  is the rank of the item  $k$  in the ordering  $r$ . For notation brevity, given two rankings,  $r$  and  $r'$ , we further set

$$\text{sign}(r(k), r(l); r'(k), r'(l)) = \text{sign}(r(k), r(l)) \text{sign}(r'(k), r'(l)).$$

We say  $r$  and  $r'$  are order consistent at  $(k, l)$  if  $\text{sign}(r(k), r(l); r'(k), r'(l)) \geq 0$ , and strict order consistent if  $\text{sign}(r(k), r(l); r'(k), r'(l)) > 0$ .

**Definition 1.** Given two full rankings  $r, r' \in \mathfrak{S}_M$ , the Kendall tau distance between them counts how many pairs of items are ordered differently in  $r$  and  $r'$ . That is,

$$d_\tau(r, r') := \frac{1}{2} \sum_{i < j} (1 - \text{sign}(r(i), r(j); r'(i), r'(j))). \quad (1)$$

### 2.1 Bayesian Stochastically Order Consistent Models

Given the set of  $M$  items, each item  $i \in [M]$  has an underlying unknown state  $\theta_i \in \Theta$ , where  $\Theta$  is a (strictly) ordered set. The vector of states  $\boldsymbol{\theta} = (\theta_1, \dots, \theta_M)$  is drawn from a prior  $\mu$ . Depending on the model,  $\boldsymbol{\theta}$  may represent either a reference ranking or underlying item costs.

For each item, each agent gets a signal from a (strictly) ordered set  $\mathcal{X}$ . Conditioned on a realized  $\boldsymbol{\theta}$ , each agent  $n \in [N]$  independently receives a private signal vector  $\mathbf{x}_n \in \mathcal{X}^M$  from a distribution  $P_n(\cdot \mid \boldsymbol{\theta})$  over these vectors of signals.

Let  $\mathbf{X} = (\mathbf{x}_n)_{n \in [N]}$  be the random matrix of all agents' signals, where each agent  $n$  receives a signal vector  $\mathbf{x}_n = (x_{n,1}, \dots, x_{n,M})$ . When realized, these values are denoted as  $\mathbf{X}$  for the matrix,  $\mathbf{x}_n$  for the row of agent  $n$ , and  $x_{n,i}$  for the entry for agent  $n$  and item  $i$ .

An *information structure* consists of a prior  $\mu$  on  $\boldsymbol{\theta}$  and the collection of conditional signal distributions  $\mathbf{P} = (P_n)_{n \in [N]}$ .

Several classical ranking models embody this setting and satisfy Bayesian SOC, including the Mallows model [19], the Thurstone model [28], and additive random utility models [28, 26, 1]. In the Mallows model, the state  $\boldsymbol{\theta} \in \mathfrak{S}_M$  is a latent reference ranking and agent  $n$  observes a noisy ranking  $\mathbf{x}_n$  with probability proportional to  $\exp(-d_\tau(\boldsymbol{\theta}, \mathbf{x}_n)/\eta_n)$ . In the Thurstone model,  $\boldsymbol{\theta} \in \mathbb{R}^M$  is a vector of latent item costs and agent  $n$  observes independent Gaussian perturbations of these costs before ranking lower observations higher. Additive random utility models replace the Gaussian noise with independent log-concave noise, covering common random-utility specifications such as Plackett–Luce [18, 21]. Heterogeneity enters through the agent-specific noise parameters or signal

distributions. The full specifications are in Appendix A.1 and Examples 1 to 3. For our results, we will need distributions to satisfy the following properties.

Given  $i, j \in [M]$  and strictly ordered sets  $\mathcal{Y}, \mathcal{Z}$ ,  $\mathbf{y} \in \mathcal{Y}^M, \mathbf{z} \in \mathcal{Z}^M$  are *order consistent* (order concordant) at  $(i, j)$  if their relative orderings at those indices match.

**Definition 2 (Kleinberg et al. [12]).** Given  $M$ , and ordered sets  $\Theta$  and  $\mathcal{X}$ , a conditional probability  $P : \Theta^M \rightarrow \Delta(\mathcal{X}^M)$  is stochastically (order) consistent if and only if for any distinct  $i, j \in [M]$ , state vector  $\boldsymbol{\theta} \in \Theta^M$ , and signal vector  $\mathbf{x} \in \mathcal{X}^M$  which are order consistent at  $(i, j)$ ,

$$P(\mathbf{x} = \mathbf{x} \mid \boldsymbol{\theta}) \geq P(\mathbf{x} = \sigma_{ij}(\mathbf{x}) \mid \boldsymbol{\theta})$$

where  $\sigma_{ij}(\mathbf{x})$  denotes the vector  $\mathbf{x}$  with the  $i$ -th and  $j$ -th components swapped.<sup>4</sup> Moreover, the inequality is strict when  $\boldsymbol{\theta}$  and  $\mathbf{x}$  are strict order consistent at  $(i, j)$ .

Intuitively, for any two items  $i$  and  $j$ , it is more probable that their order in the signal is consistent with their true order in  $\boldsymbol{\theta}$  than the reverse, regardless of the realizations of the other items.

**Definition 3 (Bayesian stochastically order-consistent).** An information structure  $(\mu, \mathbf{P})$  is **Bayesian stochastically order-consistent** (Bayesian SOC) if it satisfies:

- *Ex-ante identical:*  $\boldsymbol{\theta}$  is drawn from an exchangeable prior  $\mu$ .
- *Anonymity:* For each agent  $n$ , the conditional probability  $P_n$  is invariant to permutations of the items. For all permutations  $\sigma \in \mathfrak{S}_M$ ,  $\mathbf{x}$ , and  $\boldsymbol{\theta}$ ,  $P_n(\mathbf{x} \mid \boldsymbol{\theta}) = P_n(\sigma(\mathbf{x}) \mid \sigma(\boldsymbol{\theta}))$ .
- *Stochastic consistency:* The conditional signal distributions  $\mathbf{P}$  are stochastically (order) consistent, so the signals are more likely to be order consistent with the true underlying order.

A Bayesian SOC model is *strict* if there are no ties:  $\Pr[\mathbf{x}_{n,i} = \mathbf{x}_{n,j}] = 0$  and  $\Pr[\theta_i = \theta_j] = 0$  for all  $n \in [N]$  and  $i \neq j \in [M]$ . Finally,  $\mathbf{P}$  has full support if  $P_n(\cdot \mid \boldsymbol{\theta})$  assigns positive probability to every signal vector for all  $n$ .

Note that agents can be heterogeneous when  $P_n \neq P_{n'}$ . Throughout the paper, for readability, we use probability-mass notation such as  $P(\mathbf{x} = \mathbf{x} \mid \boldsymbol{\theta})$  and write sums over states or signals. In continuous state or signal spaces, these statements and calculations are interpreted with respect to the relevant densities and integrals under common dominating measures.

Thus, our general requirement is satisfied by several standard ranking models, rather than by a mechanism-specific prior. In Appendix F, we discuss its connection to the Independence of Irrelevant Alternatives principle and give a necessity result illustrating why the SOC orientation matters for agreement-based incentives.

## 2.2 Peer Prediction Mechanism Design for Ordinal Data

We want to elicit ordinal information from agents. For each agent  $n$ , we formalize this via a *target function*  $\Gamma_n : \mathcal{X}^M \rightarrow \mathcal{R}_n$ , which maps a signal vector to a specific (truthful) report.<sup>5</sup>

**Full ranking**  $\mathcal{R}_n = \mathfrak{S}_M$  and  $\Gamma_n = \text{ord}(\mathbf{x})$ . The target function  $\text{ord}(\mathbf{x})$  outputs a permutation  $r$  with  $r(i) < r(j)$  if and only if  $x_i < x_j$ ; lower signals are better.

<sup>4</sup> If the signal space  $\mathcal{X}^M$  is continuous or general space, we assume the conditional probability is absolutely continuous with respect to some common measure, and the inequality holds for the probability density function.

<sup>5</sup> To simplify the analysis, we assume the information structure is a strict Bayesian SOC model (see Section 2.1), which ensures distinct signal values and a unique target function.

**Top set** Given a size parameter  $K_n$ ,  $\mathcal{R}_n = \binom{[M]}{K_n}$  and  $\Gamma_n = \text{top}_{K_n}(\mathbf{x})$ . The target function outputs the indices of the  $K_n$  smallest signals.

**Subset assignment** Given an assigned subset  $S_n \subseteq [M]$ , the target function  $\Gamma_n$  is the ranking  $\text{ord}(\mathbf{x})$  restricted to these items. For instance, an assigned pair,  $|S_n| = 2$ , reproduces the setting in Chen et al. [4]; see Section 4.1.

To incentivize agents to truthfully report the target function of their private signals, a peer prediction mechanism creates a game between the agents. Formally, given target functions  $(\Gamma_n)_{n \in [N]}$ , each agent  $n$  privately observes her signal  $\mathbf{x}_n \in \mathcal{X}^M$ , and reports  $\tilde{r}_n \in \mathcal{R}_n$  which is potentially different from the truthful target report  $r_n := \Gamma_n(\mathbf{x}_n)$ . A *peer prediction mechanism*  $(U_n)_{n \in [N]}$  takes all agents' reports  $\tilde{\mathbf{r}} = (\tilde{r}_1, \dots, \tilde{r}_N)$  and pays/scores agent  $n$  with  $U_n(\tilde{\mathbf{r}}) \in \mathbb{R}$ .

Each agent  $n$ 's strategy is a random function from her signal to a report  $\alpha_n : \mathbf{x}_n \mapsto \tilde{r}_n$ , and the randomness of their strategies is independent of each other's and all signals. A (mixed) strategy profile  $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_N)$  is a collection of all agents' strategies. All agents are rational.

**Definition 4.** A mechanism is **truthful** if the truthful strategies form a Bayesian Nash equilibrium where

$$\mathbb{E}[U_n(\Gamma_n(\mathbf{x}_n), \mathbf{r}_{-n}) \mid \mathbf{x}_n] \geq \mathbb{E}[U_n(\alpha_n(\mathbf{x}_n), \mathbf{r}_{-n}) \mid \mathbf{x}_n]$$

for any alternative strategy  $\alpha_n$  where the randomness is over the peers' truthful reports  $\mathbf{r}_{-n}$ .

Beyond expected utility, we also consider a stronger property where the truthful score distribution stochastically dominates the score distribution of any deviation.

**Definition 5 ([32]).** A mechanism is **stochastically dominant truthful** if for every agent  $n$ , signal  $\mathbf{x}_n$ , and deviating strategy  $\alpha_n$ , the truthful payment random variable first-order stochastically dominates the payment from the deviation when the peers report truthfully. Formally, for all  $w \in \mathbb{R}$ :

$$\Pr[U_n(\Gamma_n(\mathbf{x}_n), \mathbf{r}_{-n}) \geq w \mid \mathbf{x}_n] \geq \Pr[U_n(\alpha_n(\mathbf{x}_n), \mathbf{r}_{-n}) \geq w \mid \mathbf{x}_n]$$

where the randomness is over the peers' truthful reports  $\mathbf{r}_{-n}$ .

Traditional truthful mechanisms require agents to be risk neutral and for their *utilities* to be linear with respect to their *scores*. With these assumptions, if the expected score decreases, so does their utility. However, these assumptions are often unrealistic. For example, students typically care about letter grades rather than numerical scores; in a crowd work setting, there may be a minimum score needed for rehiring. Stochastically dominant truthfulness (SDT) addresses this by guaranteeing that the score distribution from truth-telling first-order stochastically dominates any deviation, incentivizing truthfulness for all monotone utility functions. This robustness makes SDT mechanisms suitable for practical applications where agents' utilities may be unknown or nonlinear and where agents may not be risk neutral, such as peer assessment or human feedback for AI.

### 3 Kendall Tau Mechanisms for Eliciting Full Rankings

We introduce the Kendall tau mechanism for full rankings (Algorithm 1), which penalizes agents based on their Kendall tau distance from peers. We add the constant  $\binom{M}{2}$  to simplify the notation. Since Kendall tau distance treats a ranking as a collection of all pairwise comparisons, the mechanism rewards agents for every pair  $(i, j)$  whose order is consistent with a peer's order. Thus, we may view Kendall tau mechanisms as agreement mechanisms on pairwise comparisons.

As a warm-up, Section 3.1 proves that the Kendall tau mechanism is truthful. In particular, Lemma 1 uses symmetrization to compare the probability of order consistency and inconsistency.



**ALGORITHM 1:** Kendall Tau Mechanism for Full-Rank Data**Input:**  $\tilde{\mathbf{r}} \in \mathfrak{S}_M^N$ , the reported full rankings of  $N$  agents over  $M$  items**foreach** agent  $n \in [N]$  **do**

Compute score

$$U_n(\tilde{\mathbf{r}}) = \sum_{n' \neq n} \left( \binom{M}{2} - 2d_\tau(\tilde{r}_n, \tilde{r}_{n'}) \right) = \sum_{n' \neq n} \sum_{i < j} \text{sign}(\tilde{r}_n(i), \tilde{r}_n(j); \tilde{r}_{n'}(i), \tilde{r}_{n'}(j)). \quad (2)$$

**end**

Section 3.2 strengthens the result to stochastically dominant truthfulness. In particular, Theorem 1 implies Proposition 1. Finally, Section 3.3 shows the mechanism is strongly truthful under index-equivariant strategies. All deferred proofs are in Appendices B and C.

**3.1 Truthfulness**

**Proposition 1.** *For any strict Bayesian SOC information structure, the Kendall tau mechanism in Algorithm 1 is truthful.*

The following lemma shows that under a Bayesian SOC model, the conditional probability of any peer's signals given the agent's private signal is also stochastically order consistent.

**Lemma 1.** *If  $(\mathbf{x}, \mathbf{x}')$  are sampled from a Bayesian SOC model, for all  $i, j \in [M]$ , and  $\mathbf{x}, \mathbf{x}' \in \mathcal{X}^M$  that are strict order consistent at  $(i, j)$ ,*

$$\Pr[\mathbf{x}' = \mathbf{x}' \mid \mathbf{x} = \mathbf{x}] > \Pr[\mathbf{x}' = \sigma_{i,j}(\mathbf{x}') \mid \mathbf{x} = \mathbf{x}]$$

where  $\sigma_{i,j}$  is the permutation that swaps coordinates  $i$  and  $j$ .

Proposition 1 follows immediately from the pairwise posterior-agreement property in Lemma 1: the expected utility decomposes pairwise, and each pair is maximized by the truthful comparison. We defer the proof to Appendix B.

To analyze the full mechanism and facilitate the extensions, we define a comparison function from full rankings and rewrite Kendall tau distance.

**Definition 6 (Comparison function).** *Let  $\text{comp} : \mathfrak{S}_M \rightarrow \{-1, 0, 1\}^{M \times M}$  output  $\mathbf{D} = \text{comp}(r)$  for any ranking  $r \in \mathfrak{S}_M$  so that for all  $i, j$*

$$\mathbf{D}(i, j) = \text{sign}(r(i), r(j)).$$

We can further define the comparison function on partial orders  $R \subset [M]^2$  so that  $\mathbf{D}(i, j) = \mathbf{1}[(i, j) \in R] - \mathbf{1}[(j, i) \in R]$ .

With this notation, the Kendall tau distance between  $r$  and  $r'$  is  $d_\tau(r, r') = \frac{1}{2} \sum_{i < j} (1 - \mathbf{D}(i, j) \mathbf{D}'(i, j))$ , where  $\mathbf{D} = \text{comp}(r)$  and  $\mathbf{D}' = \text{comp}(r')$ , and the score in Eq. (2) becomes

$$U_n(\tilde{\mathbf{r}}) = \sum_{n' \neq n} \sum_{i < j} \tilde{\mathbf{D}}_n(i, j) \tilde{\mathbf{D}}_{n'}(i, j) \quad (3)$$

where  $\tilde{\mathbf{D}}_n := \text{comp}(\tilde{r}_n)$  for all  $n$ . Similarly, define  $\mathbf{D}_n := \text{comp}(r_n)$  for the truthful report of agent  $n$ .

### 3.2 Stochastically Dominant Truthfulness

**Theorem 1.** *For any strict Bayesian SOC information structure, the Kendall tau mechanism in Algorithm 1 is stochastically dominant truthful.*

*Proof Overview* To prove Theorem 1, we show that if all other agents are truthful, deviating to any nontruthful report  $\tilde{r}$  yields a first-order stochastically dominated score. Specifically, given agent  $n$  and her private signal  $\mathbf{x}_n$ , we construct a sequence of reports  $(r^t)_{t=0,\dots,T}$  for agent  $n$  with payments  $U^t = U_n(r^t, \Gamma(\mathbf{X}_{-n}))$  so that  $r^0 = \tilde{r}$ ,  $r^T = \Gamma_n(\mathbf{x}_n)$  is the truthful report, and

$$U^t \geq_{\text{FOSD}} U^{t-1} \quad \text{for all } t = 1, \dots, T,$$

which completes the proof by transitivity of first-order stochastic dominance. We construct the sequence by bubble sort: at step  $t$  we pick a pair of items  $(i_t, j_t)$  that are adjacent in  $r^{t-1}$  but out of order relative to the signal,  $x_{n,i_t} < x_{n,j_t}$  but  $r^{t-1}$  ranks  $j_t$  above  $i_t$ , and we let  $r^t$  swap their ranks. Each swap is one iteration of bubble sort, so the sequence terminates at the truthful report in finite steps.

Because  $i_t$  and  $j_t$  are adjacent in both  $r^{t-1}$  and  $r^t$ , the only comparison that changes is the one between  $i_t$  and  $j_t$ . We therefore can decompose the peers' signals into *magnitude*  $\rho$ , which specifies everything except how each peer orders this pair, and *alignment*  $\mathbf{a} \in \{-1, 1\}^{N-1}$ , where  $a_{n'} = 1$  if peer  $n'$  orders  $(i_t, j_t)$  consistent with  $r^t$ . Lemma 2 shows that, conditional on a fixed magnitude, the score difference depends on the alignment only through the *margin*  $L(\mathbf{a}) = \sum_{n' \neq n} a_{n'}$ . Let  $Q(\rho, L_B = d)$  denote the conditional probability, given  $\mathbf{x}_n$ , of magnitude  $\rho$  and margin  $d$  on  $B$ . Lemma 3 then shows that the conditional distribution of margin  $L_B$  is more likely to be positive than its negation.

**Lemma 2.** *For each adjacent correction step, fixing the magnitude  $\rho$ , the score difference is determined by the peer-alignment margin. In particular, for any alignment vectors  $\mathbf{a}, \mathbf{a}'$ ,*

$$U^t(\rho, \mathbf{a}) - U^t(\rho, \mathbf{a}') = L(\mathbf{a}) - L(\mathbf{a}') \quad (4)$$

and

$$U^t(\rho, \mathbf{a}) = U^{t-1}(\rho, \mathbf{a}) + 2L(\mathbf{a}). \quad (5)$$

**Lemma 3.** *For any agent signal  $\mathbf{x}_n$ , magnitude  $\rho$ , subset  $B \subseteq [N-1]$ , and  $d > 0$ ,*

$$Q(\rho, L_B = d) \geq Q(\rho, L_B = -d), \quad (6)$$

and the inequality is strict for some  $d$ .

By Lemma 2, each correcting swap changes the score by a monotone function of the margin. By Lemma 3, positive margins dominate negative margins under Bayesian SOC. Pairing positive and negative margins gives one-step stochastic dominance, and composing the correction path proves Theorem 1. The full proof, including the magnitude/alignment definitions and the symmetric-polynomial argument for Lemma 3, is deferred to Appendix B.

### 3.3 Strong Truthfulness under Index-Equivariant Strategies

The incentive guarantee in Theorem 1 is unilateral around truthful peer reports: when peers report truthfully, an agent's truthful score distribution first-order stochastically dominates any deviation. It does not rule out other coordinated uninformative profiles. For example, if all agents report the



same fixed ranking regardless of their signals, the agreement score can be high and no single agent may want to deviate. Such coordination, however, requires a common indexing of the items, which the designer can destroy by presenting the items to each agent under random relabeling (in RLHF, by randomly shuffling and rephrasing the responses shown to each annotator). Under randomized indexing, it is without loss of generality to restrict attention to *index-equivariant* strategies. Similar notion is also proposed in [33,31].

**Definition 7.** A (mixed) strategy  $\alpha_n : \mathcal{X}^M \rightarrow \Delta(\mathfrak{S}_M)$  is **index-equivariant** if  $\alpha_n(\sigma(\mathbf{x}_n)) \stackrel{d}{=} \sigma(\alpha_n(\mathbf{x}_n))$  for all permutations  $\sigma \in \mathfrak{S}_M$  and  $\mathbf{x}_n \in \mathcal{X}^M$ , where  $\sigma$  acts on signal vectors by  $\sigma(\mathbf{x})_i := x_{\sigma^{-1}(i)}$  and on rankings by  $\sigma(r)(i) := r(\sigma^{-1}(i))$ .

Truth-telling  $\Gamma_n = \text{ord}$  is index-equivariant, as is its composition with the rank reversal  $\text{rev} \circ r(i) := M + 1 - r(i)$ , but a constant strategy is not.

**Theorem 2 (Strong truthfulness).** For any strict, full-support Bayesian SOC information structure, truth-telling attains the highest ex-ante expected payment for all agents under index-equivariant strategy profiles in Algorithm 1, and equality holds only for the truthful profile and its reversal,  $\alpha_n = \text{rev} \circ \Gamma_n$  for all  $n$ .

We defer the proof to Appendix C and sketch it here. For a fixed pair  $(i, j)$ , we define the alignment factor  $\kappa_n(\mathbf{x}) := 1 - 2 \Pr[\tilde{\mathbf{D}}_n(i, j) \neq \text{sign}(x_i, x_j) \mid \mathbf{x}_n = \mathbf{x}] \in [-1, 1]$  where  $\kappa_n = 1$  for truth-telling. With Lemma 1, we show that the expected score can be decomposed as positive terms scaled by the product  $\kappa_n \kappa_{n'} \in [-1, 1]$ , so truth-telling attains the maximum.

For the equality case, full support forces each  $\kappa_n$  to be a constant sign  $\epsilon^{ij} \in \{\pm 1\}$  identical for all agents, and equivariance ensures the flip events across item pairs. Hence every agent either reports truthfully on all pairs ( $\epsilon = 1$ ) or flips all pairs ( $\epsilon = -1$ ), the global reversal.

## 4 Extensions

### 4.1 Subset Assignments

In many practical applications, the number of items  $M$  can be too large for every agent to rank. We therefore allow each agent  $n$  to report only on an assigned subset  $S_n \subseteq [M]$  and score only common item pairs shared with peers. For the comparison matrix  $\tilde{\mathbf{D}}_n$  induced by agent  $n$ 's report, the generalized score is

$$U_n(\tilde{\mathbf{r}}) = \sum_{n' \neq n} \sum_{i, j \in S_n \cap S_{n'} : i < j} \tilde{\mathbf{D}}_n(i, j) \tilde{\mathbf{D}}_{n'}(i, j). \quad (7)$$

This formula covers both subset rankings, where  $\tilde{\mathbf{D}}_n$  orders all items in  $S_n$ , and subset top-sets, where  $\tilde{\mathbf{D}}_n$  records whether one item is inside the reported top set within  $S_n$  and the other is outside. The key feasibility condition is *pairwise coverage*: every pair that appears in one agent's assigned subset must also appear in at least one other agent's subset. Under this condition, every reported comparison can be checked by some peer, and the same correction-path argument applies with margins restricted to these active peers. The proof is in Appendix D.

**Theorem 3.** For any strict Bayesian SOC information structure and pairwise coverage assignment  $\{S_n\}_{n=1}^N$ , the generalized Kendall tau mechanism scoring agreement only on shared pairs is stochastically dominant truthful for subset rankings and subset top-sets.

*Random assignments.* Theorem 3 also applies to randomized assignments. If  $\{S_n\}_{n=1}^N$  is sampled independently of signals and reports, then conditional on any realized assignment satisfying pairwise coverage, the mechanism is stochastically dominant truthful. Since first-order stochastic dominance is preserved under mixtures, the ex-ante mechanism is also stochastically dominant truthful whenever pairwise coverage holds almost surely. If coverage can fail, the guarantee applies only to covered comparisons; uncovered comparisons are not checked by any peer and cannot be incentivized by peer agreement alone.

## 4.2 General Bucket Orders

Bucket orders generalize full rankings, top- $K$  reports, and intermediate ordinal reports such as accept/borderline/reject tiers or ranked shortlists with unordered remainders.

**Definition 8 (Bucket order [5]).** A bucket order of shape  $\mathbf{K} = (K_1, \dots, K_L)$  with  $\sum_{\ell=1}^L K_\ell = M$ ,  $L \geq 2$ , and  $K_\ell > 0$  for all  $\ell$  is an ordered partition of  $[M]$  into blocks (buckets)  $B_1, \dots, B_L$  with  $|B_\ell| = K_\ell$ . We encode a report  $r$  as  $\beta_r(i) \in [L]$  for the bucket of item  $i$  under  $r$ , with  $\beta_r(i) = 1$  denoting the best bucket, and its comparison matrix  $\mathbf{D}_r = \text{comp}(r)$  as

$$\mathbf{D}_r(i, j) = \text{sign}(\beta_r(i), \beta_r(j)) \in \{-1, 0, 1\},$$

so  $\mathbf{D}_r(i, j) = +1$  if  $i$  is in a strictly better bucket than  $j$ , and  $\mathbf{D}_r(i, j) = 0$  if  $i, j$  share a bucket.

In particular, full rankings have shape  $(1, \dots, 1)$ , and top- $K$  reports have bucket shape  $(K, M - K)$ . We can also define median-pivot report that identifies the median item and separately reports the items above and below of shape  $(\lfloor \frac{M-1}{2} \rfloor, 1, \lfloor \frac{M-1}{2} \rfloor)$  for odd  $M$ .

We can generalize our Kendall tau mechanism to bucket orders. The target function  $\Gamma_n(\mathbf{x}_n)$  sorts the signal and assigns the  $K_1$  items with the smallest (best) signals to  $B_1$ , the next  $K_2$  to  $B_2$ , and so on. We allow agents to have different shapes  $\mathbf{K}^n = (K_1^n, \dots, K_{L_n}^n)$  with  $B_\ell^n$  for agent  $n$ 's buckets. For comparison matrices, write  $\langle \mathbf{A}, \mathbf{B} \rangle := \sum_{i < j} \mathbf{A}(i, j) \mathbf{B}(i, j)$ . The Kendall tau mechanism pays agent  $n$

$$U_n(\tilde{\mathbf{r}}) = \sum_{n' \neq n} \langle \mathbf{D}_{\tilde{\mathbf{r}}_n}, \mathbf{D}_{\tilde{\mathbf{r}}_{n'}} \rangle. \quad (8)$$

**Theorem 4.** For any strict Bayesian SOC information structure and any per-agent shapes  $\mathbf{K}^n$ , the Kendall tau mechanism in Eq. (8) restricted to shape- $\mathbf{K}^n$  reports is stochastically dominant truthful.

The proof mirrors the full-ranking argument but lets a correcting step move two items across adjacent bucket boundaries rather than adjacent ranks. Only comparisons involving the swapped items and the two affected buckets can change. For each peer, the score increment has the form  $c_{t,n'} a_{n'}$ , where  $a_{n'} \in \{-1, 1\}$  records alignment with the corrected pair and  $c_{t,n'} \geq 0$  is determined by the peer's bucket placement. Thus the one-step score improvement is a nonnegative weighted margin  $\langle \mathbf{c}_t, \mathbf{a} \rangle$ . Bayesian SOC makes aligned peer configurations dominate anti-aligned ones conditional on the magnitude, yielding one-step stochastic dominance; composing the bucket-correction path proves the theorem. Full rankings have shape  $1^M$ , and top sets have shape  $(K, M - K)$ .

This result should be read as a positive result for bucket orders, not as a claim that all partial-order targets are stochastically dominant elicitable by pairwise agreement. More general partial-order targets may introduce dependencies among elicited comparisons and may require different arguments.

The bucket-order theorem implies the top-set result. For top sets  $r, r' \subseteq [M]$  with comparison matrices  $\mathbf{D}, \mathbf{D}'$ , the specialized score has the overlap identity

$$\sum_{i < j} \mathbf{D}(i, j) \mathbf{D}'(i, j) = M|r \cap r'| - |r||r'|.$$

Thus, for fixed reported set sizes, the mechanism rewards overlap with peers' top sets. Equivalently, if a top set is encoded as a binary inclusion vector, this score is an affine transformation of the Enforced Agreement (EA) score [32] on binary reports; see Proposition 2.

**Corollary 1 (Top-set elicitation).** *For any strict Bayesian SOC information structure and fixed top-set sizes  $(K_n)_{n \in [N]}$ , the Kendall tau mechanism is stochastically dominant truthful for top- $K_n$  reports.*

Corollary 1 follows as a special case of the more general bucket-order theorem; the specialized score identity and its relation to Enforced Agreement are recorded in Appendix E.1.

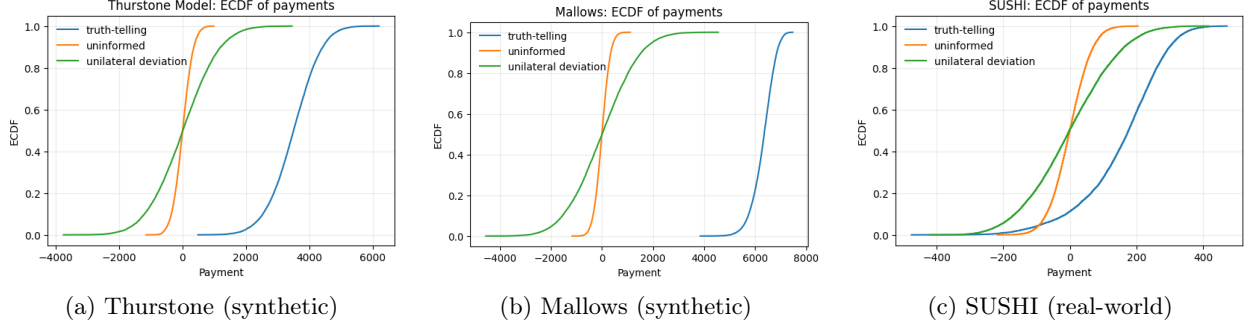
## 5 Experiments

### 5.1 Agent Payments Comparison

We empirically evaluate the Kendall–tau peer-prediction mechanism by comparing empirical payment distributions under truthful reporting, uninformed random reporting, and unilateral random deviation. In each trial, agents submit full rankings and payments are computed using the score in Eq. (2). For the synthetic experiments we use  $N = 25$  agents,  $M = 30$  items, and 600 independent trials. In the Thurstone experiment, item costs are drawn i.i.d. from a standard Gaussian distribution, and each agent observes noisy costs before ranking smaller observations higher. In the Mallows experiment, a latent reference ranking is sampled and agents' truthful rankings are noisy Mallows perturbations of this reference with dispersion parameter  $\eta_n = 0.8$ . For SUSHI [11], we treat the observed human ranking as the truthful/observed report: each trial samples  $N = 25$  observed human rankings over  $M = 10$  sushi items. This real-data setting is meant as an empirical stress test rather than a model guaranteed to satisfy Bayesian SOC.

For each model or dataset, we compare three reporting regimes. In *truth-telling*, every agent reports the ranking induced by their signal; for SUSHI, this means the observed human ranking. In *uninformed reporting*, every report is an independent uniformly random ranking. In *unilateral deviation*, one focal agent reports a uniformly random ranking while peers report truthfully or by their observed SUSHI rankings; repeating over agents and trials gives the deviation distribution. The plots show empirical cumulative distribution functions (ECDFs), so a lower, right-shifted curve corresponds to higher payments. Thus stochastic dominance of truthful reporting means the truthful ECDF lies weakly below the deviation ECDF at every threshold. Additional implementation details are in Appendix G.

Across the synthetic Bayesian SOC models (Thurstone and Mallows), the empirical payment distributions closely match the theoretical prediction of stochastically dominant truthfulness, with the ECDF of truthful reporting lying uniformly below, or equivalently right-shifted relative to, the alternatives in our simulations (Figure 1). On the real-world SUSHI dataset, while the dominance does not hold everywhere, the observed-report regime nevertheless yields a consistently right-shifted score distribution with substantially more mass on higher payments (Figure 1). Taken together, these results support the distributional incentive prediction in synthetic SOC models and show favorable observed-report score distributions on heterogeneous, nonparametric preference data. In Appendix G, we repeat the comparison against structured unilateral deviations, including near-truthful



**Fig. 1.** ECDFs of agent payments under three reporting regimes: truth-telling (observed human rankings for SUSHI), uninformed reporting (uniform random permutations), and unilateral deviation (one agent reports a random permutation while others report truthfully or by their observed SUSHI rankings).

perturbations and coordination-style reports; the synthetic results remain favorable to truthful reporting, and the SUSHI observed-report distribution is generally the most favorable in this real-data stress test.

For SUSHI, where no signal-noise or effort parameter is observed, we also degrade a focal agent’s observed ranking by applying  $k$  random adjacent transpositions while all peers keep their observed rankings, and then record the focal Kendall tau payment. We use the same SUSHI sampling protocol as above and sweep  $k \in \{0, 1, 2, 3, 5, 8, 13, 21, 34, 45\}$ , where  $k = 0$  is the observed ranking and  $k = 45 = \binom{10}{2}$  approaches a random report for  $M = 10$ . This degradation test is shown in Figure 2(a); full implementation details are in Appendix G.

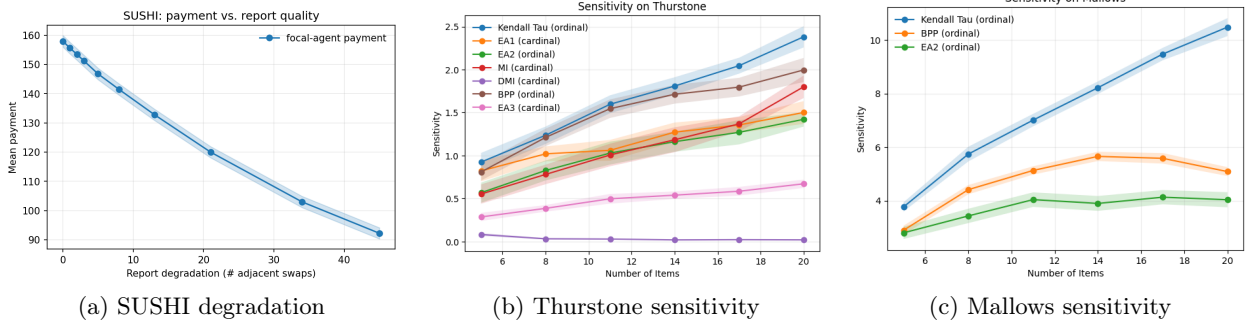
## 5.2 Sensitivity to Report Quality

For the synthetic SOC models, the ECDF comparison above directly tests the distributional incentive prediction. The following sensitivity experiment is different: it is not another truthfulness test, but an ex-ante measurement-quality test asking whether the payment is locally responsive to the quality of an agent’s signal. In the synthetic Thurstone and Mallows models, we designate one focal agent while all other agents report truthfully, and we vary the focal agent’s noise or dispersion parameter  $\eta_n$ . Thus  $\eta_n$  does not parameterize a strategic report deviation; in the Thurstone plot, it denotes the focal agent’s signal-noise variance, so larger  $\eta_n$  corresponds to noisier signals or lower effective effort. Let  $U_n(\eta_n)$  denote the focal agent’s payment. Following [30], we measure normalized sensitivity by

$$\text{Sens}_n = \frac{-\frac{d}{d\eta_n} \mathbb{E}[U_n(\eta_n)]}{\sigma[U_n(\eta_n)]},$$

so larger values mean that expected payment falls more sharply, relative to payoff variability, as the focal agent’s signal becomes noisier. The Thurstone panel compares Kendall–tau scoring with six baselines, including cardinal baselines that use numerical signals and ordinal baselines that use only rankings; the Mallows panel reports the two ordinal baselines shown there, EA2 and BPP.

On SUSHI, mean payment decreases as the observed report is progressively degraded. As the number of items grows, the Kendall–tau mechanism has the largest normalized sensitivity among the tested baselines in both synthetic models (Figure 2). Thus, in these settings, the guarantee does not appear to reduce local sensitivity, and the score also tracks report quality on real preference data.



**Fig. 2.** Measurement-quality tests. Panel (a) degrades a focal observed SUSHI ranking by random adjacent swaps and plots mean focal-agent payment with a 95% confidence band. Panels (b)–(c) compare normalized sensitivity for Kendall tau and baseline mechanisms under the Thurstone and Mallows models.

## 6 Discussion and Future Work

We introduce a simple peer-prediction mechanism that rewards pairwise agreement using Kendall tau, giving a general framework for ordinal information elicitation. Under Bayesian Stochastically Order Consistent (SOC) information structures—a broad class including Mallows and Thurstone models—truthful reporting stochastically dominates any deviation. Unlike mechanisms that rely on learning or strong prior assumptions, our mechanism handles heterogeneous agents with complex, continuous signal spaces. It also elicits fixed-shape bucket orders, including top- $K$  sets, and handles subset assignments. Experiments on synthetic data and SUSHI preferences suggest that truthful or observed reporting receives more favorable score distributions than uninformed or deviating strategies in the tested settings.

While our mechanism has these advantages, there are also limitations. The largest is our assumption of Stochastically Order Consistent (Bayesian SOC), which is satisfied by standard ranking distributions but might not always hold. One example is the case of “cheap signals” [9,15]. Here agents have additional information on which they can coordinate besides the quality of the items that they are ranking. For example, instead of ranking peer reviews according to their overall quality, agents could rank them according to the institutions of the authors. This is a known issue in the peer-review literature, and our mechanism would not work as well in these cases. Our index-equivariance assumption in Definition 7 excludes this setting.

Another limitation occurs when not every agent evaluates every task. We can only reward an ordering of items  $i$  and  $j$  if another agent also orders them. This still allows flexible assignments—not every agent needs every task—but other mechanisms allow even more general assignments [23,4].

Future research may investigate generalizations of the Bayesian SOC assumption, exploring whether our mechanism extends to alternative conditions such as Multivariate Total Positivity of order 2 [6].

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## Appendix

### A Preliminaries and Model Examples

A note on notation: as in the main text, we write probabilities and sums for readability. In continuous state or signal spaces, the corresponding statements are interpreted using densities and integrals with respect to common dominating measures.

A (strict) *ordered set*  $\mathcal{Y}$  has a strict total order so that for all  $y, y', y'' \in \mathcal{Y}$

1. If  $y < y'$ , then not  $y' < y$  (asymmetric)
2. If  $y < y'$ ,  $y' < y''$ , then  $y < y''$  (transitive)
3. If  $y \neq y'$ , then  $y < y'$  or  $y' < y$  (complete)

A (strict) *partially ordered set* relaxes completeness.

#### A.1 Standard Bayesian SOC Models

*Example 1 (Mallows model).* Here the vector of item states  $\boldsymbol{\theta}$  and agent  $n$ 's signal vector  $\mathbf{x}_n$  are each rankings of the items. The signal for each item is then its order in the ranking, so  $\Theta = \mathcal{X} = [M]$ . Because the full signal is a permutation, the vectors lie in the smaller set  $\mathfrak{S}_M \subset \Theta^M = \mathcal{X}^M$ . For the *Mallows model* with  $\boldsymbol{\eta}, \boldsymbol{\theta} \in \mathfrak{S}_M$  is a reference ranking (e.g., the ground-truth ordering of items) and  $\boldsymbol{\eta} = (\eta_1, \dots, \eta_N) \in \mathbb{R}_{>0}^N$  is a noise parameter vector, allowing each agent to have their own “ranking noise.”

In this model, agent  $n$ 's observed ranking  $\mathbf{x} \in \mathfrak{S}_M$  is generated with probability

$$P_n(\mathbf{x}_n = \mathbf{x} \mid \boldsymbol{\theta}) \propto \exp\left(-\frac{1}{\eta_n} d_\tau(\boldsymbol{\theta}, \mathbf{x})\right)$$

where  $d_\tau$  is Kendall's tau distance (see Definition 1), which measures the number of pairwise disagreements between two rankings.

In the Mallows model, smaller  $\eta_n$  yields rankings closer to the reference  $\boldsymbol{\theta}$ , while larger  $\eta_n$  produces noisier (more random) rankings.

*Example 2 (Thurstone model).* Let  $\Theta = \mathcal{X} = \mathbb{R}$ . The latent state  $\boldsymbol{\theta}$  is a vector in  $\mathbb{R}^M$ , each coordinate  $\theta_i$  being the (true) cost of item  $i$ . We assume  $\boldsymbol{\theta}$  is drawn from a fixed non-atomic distribution (for example, a standard Gaussian).

Each agent  $n$  then sees a noisy version of each  $\theta_i$ : they observe Gaussian signals  $\mathbf{x}_{n,i}$ , where

$$\mathbf{x}_{n,i} \sim \mathcal{N}(\theta_i, \eta_n)$$

with  $\eta_n > 0$  specifying the noise level for agent  $n$ . All noise is independent across items and agents.

In the Thurstone model, each agent sees noisy measurements of the underlying item costs, with more reliable agents having lower  $\eta_n$ . The Thurstone model is a classical foundation for statistical models of ranking and paired comparison. While agents ultimately report an ordering (ranking) derived from these perturbed costs, unlike the Mallows model, in the Thurstone model the agent signal is not a ranking per se.

*Example 3 (Additive RUM).* Let  $\Theta = \mathcal{X} = \mathbb{R}$ . Here, the state vector  $\boldsymbol{\theta} \in \mathbb{R}^M$  contains the true costs of the  $M$  items, drawn from an exchangeable prior  $\mu$ . Each agent  $n$  observes, for each item  $i$ , a noisy signal:

$$x_{n,i} = \theta_i + \epsilon_{n,i}$$

where the noise terms  $\epsilon_{n,i}$  are independent across agents and items, and drawn from a fixed log-concave distribution (e.g., Gaussian, Laplacian, uniform, or Gumbel—the last of which gives the Plackett–Luce ranking model [18, ?]).

This is a general model of how agents’ perceptions of item costs are “blurred” by independent noise. The requirement that the noise be log-concave ensures technical properties. This family includes most “random utility” models popular in social choice and statistics. The Thurstone model is a special case of the additive RUM where the noise on each item is Gaussian and independent across both items and agents.

## B Full-Ranking Proofs and Details for Theorem 1

*Proof (Proof of Lemma 1).* It suffices to show that  $\Pr[\mathbf{x}' = \mathbf{x}', \mathbf{x} = \mathbf{x}] > \Pr[\mathbf{x}' = \sigma(\mathbf{x}'), \mathbf{x} = \mathbf{x}] > 0$ . By the definition of a Bayesian SOC model, there exists a prior  $\mu$  and conditional probabilities  $P$  and  $P'$  so that

$$\begin{aligned} & \Pr[\mathbf{x}' = \mathbf{x}', \mathbf{x} = \mathbf{x}] \\ &= \sum_{\boldsymbol{\theta}} P'(\mathbf{x}'|\boldsymbol{\theta})P(\mathbf{x}|\boldsymbol{\theta})\mu(\boldsymbol{\theta}) \\ &= \frac{1}{2} \left( \sum_{\boldsymbol{\theta}} P'(\mathbf{x}'|\boldsymbol{\theta})P(\mathbf{x}|\boldsymbol{\theta})\mu(\boldsymbol{\theta}) + \sum_{\boldsymbol{\theta}} P'(\mathbf{x}'|\sigma(\boldsymbol{\theta}))P(\mathbf{x}|\sigma(\boldsymbol{\theta}))\mu(\sigma(\boldsymbol{\theta})) \right) \quad (\text{add twice}) \\ &= \frac{1}{2} \left( \sum_{\boldsymbol{\theta}} P'(\mathbf{x}'|\boldsymbol{\theta})P(\mathbf{x}|\boldsymbol{\theta})\mu(\boldsymbol{\theta}) + \sum_{\boldsymbol{\theta}} P'(\mathbf{x}'|\sigma(\boldsymbol{\theta}))P(\mathbf{x}|\sigma(\boldsymbol{\theta}))\mu(\boldsymbol{\theta}) \right) \quad (\mu \text{ exchangeable}) \\ &= \frac{1}{2} \sum_{\boldsymbol{\theta}} (P'(\mathbf{x}'|\boldsymbol{\theta})P(\mathbf{x}|\boldsymbol{\theta}) + P'(\sigma(\mathbf{x}')|\boldsymbol{\theta})P(\sigma(\mathbf{x})|\boldsymbol{\theta})) \mu(\boldsymbol{\theta}) \quad (\text{anonymous and } \sigma(\sigma(\boldsymbol{\theta})) = \boldsymbol{\theta}) \end{aligned}$$

Therefore, the difference is

$$\begin{aligned} & \Pr[\mathbf{x}' = \mathbf{x}', \mathbf{x} = \mathbf{x}] - \Pr[\mathbf{x}' = \sigma(\mathbf{x}'), \mathbf{x} = \mathbf{x}] \\ &= \frac{1}{2} \sum_{\boldsymbol{\theta}} (P'(\mathbf{x}'|\boldsymbol{\theta})P(\mathbf{x}|\boldsymbol{\theta}) + P'(\sigma(\mathbf{x}')|\boldsymbol{\theta})P(\sigma(\mathbf{x})|\boldsymbol{\theta}) \\ & \quad - P'(\sigma(\mathbf{x}')|\boldsymbol{\theta})P(\mathbf{x}|\boldsymbol{\theta}) - P'(\mathbf{x}'|\boldsymbol{\theta})P(\sigma(\mathbf{x})|\boldsymbol{\theta})) \mu(\boldsymbol{\theta}) \\ &= \frac{1}{2} \sum_{\boldsymbol{\theta}} (P'(\mathbf{x}'|\boldsymbol{\theta}) - P'(\sigma(\mathbf{x}')|\boldsymbol{\theta})) (P(\mathbf{x}|\boldsymbol{\theta}) - P(\sigma(\mathbf{x})|\boldsymbol{\theta})) \mu(\boldsymbol{\theta}) \end{aligned}$$

which depends on

$$(P'(\mathbf{x}'|\boldsymbol{\theta}) - P'(\sigma(\mathbf{x}')|\boldsymbol{\theta})) (P(\mathbf{x}|\boldsymbol{\theta}) - P(\sigma(\mathbf{x})|\boldsymbol{\theta})). \quad (9)$$

If  $\boldsymbol{\theta}$  is order consistent with  $\mathbf{x}$  at  $(i, j)$ , and thus consistent with  $\mathbf{x}'$ , then by Definition 2  $P'(\mathbf{x}'|\boldsymbol{\theta}) > P'(\sigma(\mathbf{x}')|\boldsymbol{\theta})$  and  $P(\mathbf{x}|\boldsymbol{\theta}) > P(\sigma(\mathbf{x})|\boldsymbol{\theta})$ , so Eq. (9) is positive. Similarly, if  $\boldsymbol{\theta}$  is order consistent with  $\sigma(\mathbf{x})$ , Eq. (9) is also positive. Therefore,  $\Pr[\mathbf{x}' = \mathbf{x}', \mathbf{x} = \mathbf{x}] > \Pr[\mathbf{x}' = \sigma(\mathbf{x}'), \mathbf{x} = \mathbf{x}]$ , which completes the proof.

*Proof (Proof of Proposition 1).* By Definition 6, the mechanism depends only on pairwise comparisons. Hence, we can extend the Kendall tau mechanism to take comparison matrices as inputs rather than rankings. We show that this extended mechanism maintains truthfulness even when agents report arbitrary comparison matrices that do not correspond to any valid ranking.

Consider agent  $n$ . If all other agents report truthfully, we need to show that truth-telling maximizes the agent's score. By Eq. (3), the agent's best response consists of choosing  $\mathbf{D}(i, j)$  for each pair  $(i, j)$  separately to maximize her score. For any  $(i, j)$  with  $x_{n,i} < x_{n,j}$ , the score of truth-telling on pair  $(i, j)$  with agent  $n'$  is

$$\mathbb{E}[\mathbf{D}_{n'}(i, j) \mid \mathbf{x}_n] = \Pr[x_{n',i} < x_{n',j} \mid \mathbf{x}_n] - \Pr[x_{n',i} > x_{n',j} \mid \mathbf{x}_n] > 0$$

by Lemma 1, and it is larger than the score of reporting the wrong comparison  $-\mathbb{E}[\mathbf{D}_{n'}(i, j) \mid \mathbf{x}_n]$ . The case  $x_i > x_j$  holds by symmetry. Summing over all peers  $n'$  and pairs  $(i, j)$ , the unique best response is the comparison matrix consistent with the signal  $\mathbf{x}_n$ , corresponding to the truthful ranking  $\text{ord}(\mathbf{x}_n) = \Gamma_n(\mathbf{x}_n)$ .

**Definition 9 (Correction path).** *Given an agent's signal  $\mathbf{x}$  and a reported ranking  $\tilde{r}$ , we define the correction path as the sequence of rankings  $r^0, r^1, \dots, r^T$  to transform  $\tilde{r}$  into  $\text{ord}(\mathbf{x})$  through the bubble-sort algorithm.*

1. Start:  $r^0 = \tilde{r}$ .
2. Step  $t$ : Given  $r^{t-1}$ , we find a rank  $k_t$  such that the item at rank  $k_t + 1$  (called  $i_t$ ) and the item at rank  $k_t$  (called  $j_t$ ) satisfy  $x_{i_t} < x_{j_t}$ —out of order relative to the signal. We then swap their ranks, setting  $r^t(i_t) = k_t$  and  $r^t(j_t) = k_t + 1$ .
3. End:  $r^T = \text{ord}(\mathbf{x})$ , where no such adjacent contradictions exist.

Finally, let  $U^t(\mathbf{X}_{-n}) = U_n(r^t, \Gamma(\mathbf{X}_{-n}))$  denote agent  $n$ 's payment at step  $t$  given peer signals. Note that  $T = O(M^2)$  is bounded, because bubble sort performs at most  $O(M^2)$  swaps.

**Definition 10 (Magnitude, alignment vectors, and margins).** *Given an item pair  $(i_t, j_t)$  and a reference agent  $n$  with signal vector  $\mathbf{x}_n$  so that  $x_{n,i_t} < x_{n,j_t}$ , we classify the outcome space of peers' signal matrix  $\mathbf{X}_{-n} \in \mathcal{X}^{(N-1) \times M}$  into equivalence classes  $\mathcal{V}$  called magnitude.<sup>6</sup> We say two matrices  $\mathbf{X}_{-n}$  and  $\mathbf{X}'_{-n}$  have the same magnitude if for every peer  $n' \in [N - 1]$ :*

1. The set of entries at the pair indices is identical:  $\{x_{n',i_t}, x_{n',j_t}\} = \{x'_{n',i_t}, x'_{n',j_t}\}$ ;
2. All other entries are identical:  $x_{n',k} = x'_{n',k}$  for all  $k \notin \{i_t, j_t\}$ .

For any  $\rho \in \mathcal{V}$ , let  $\hat{\mathbf{X}}_{-n}$  be the canonical element so that each peer's signal  $\hat{\mathbf{x}}_{n'}$  and  $\mathbf{x}_n$  are order consistent at  $(i_t, j_t)$ , with  $\hat{x}_{n',i_t} < \hat{x}_{n',j_t}$ . For any matrix  $\mathbf{X}_{-n} \in \rho$ , we define the alignment vector  $\mathbf{a} \in \{1, -1\}^{N-1}$  to indicate whether each peer is order consistent with agent  $n$ . For all  $n' \in [N - 1]$ ,

$$a_{n'} = \text{sign}(x_{n',i_t}, x_{n',j_t}; x_{n,i_t}, x_{n,j_t}).$$

Given  $\rho$  and  $\mathbf{a}$  with canonical element  $\hat{\mathbf{X}}_{-n}$ ,

$$x_{n',i} = \begin{cases} \hat{x}_{n',j_t} & \text{if } i = i_t \text{ and } a_{n'} = -1 \\ \hat{x}_{n',i_t} & \text{if } i = j_t \text{ and } a_{n'} = -1 \\ \hat{x}_{n',i} & \text{otherwise.} \end{cases} \quad (10)$$

<sup>6</sup> Without loss of generality, we can re-index agents so that the set of peers is indexed by  $[N - 1]$ .

We denote the magnitude and alignment vector of a matrix by  $\rho(\mathbf{X}_{-n})$  and  $\mathbf{a}(\mathbf{X}_{-n})$ . We define  $\check{\mathbf{X}}_{-n}$  as the fully reversed configuration corresponding to  $\mathbf{a} = -\mathbf{1}$ .

Finally, for any subset of peers  $B \subseteq [N-1]$ , the margin is  $L_B(\mathbf{a}) := \sum_{n' \in B} a_{n'}$ , and  $L(\mathbf{a}) := L_{[N-1]}(\mathbf{a})$ .

Given the above notation, define

$$Q(\rho, \mathbf{a}) \triangleq \Pr[\rho = \rho, \mathbf{a} = \mathbf{a} \mid \mathbf{x}_n],$$

and use  $Q(\rho, L_B = d)$  to denote the probability of class  $\rho$  and margin  $L_B(\mathbf{a}) = d$ . Finally, let  $U^t(\rho, \mathbf{a}) = U_n(r^t, \Gamma(\mathbf{X}_{-n}))$  denote agent  $n$ 's payment at step  $t$  given peer signals parameterized by  $(\rho, \mathbf{a})$ .

*Proof (Proof of Lemma 2).* We can decompose the score into three parts.

$$\begin{aligned} U^t(\rho, \mathbf{a}) &= U^t(\mathbf{X}_{-n}) \\ &= \sum_{n'} \left( \sum_{i,j \notin \{i_t, j_t\}} \text{sign}(x_{n',i}, x_{n',j}; r_i^t, r_j^t) \right. \\ &\quad \left. + \sum_{i,j: |\{i,j\} \cap \{i_t, j_t\}|=1} \text{sign}(x_{n',i}, x_{n',j}; r_i^t, r_j^t) \right) + L(\mathbf{a}), \end{aligned}$$

because

$$L(\mathbf{a}) = \sum_{n'} \text{sign}(x_{n',i_t}, x_{n',j_t}; x_{n,i_t}, x_{n,j_t}),$$

and

$$\text{sign}(x_{n',i_t}, x_{n',j_t}; x_{n,i_t}, x_{n,j_t}) = \text{sign}(x_{n',i_t}, x_{n',j_t}; r_i^t, r_j^t).$$

For the difference  $U^t(\rho, \mathbf{a}) - U^t(\rho, \mathbf{a}')$ , the first terms agree because of the same class. Hence

$$\begin{aligned} U^t(\rho, \mathbf{a}) - U^t(\rho, \mathbf{a}') &= U^t(\mathbf{X}_{-n}) - U^t(\mathbf{X}'_{-n}) \\ &= \sum_{n'} \sum_{i,j: |\{i,j\} \cap \{i_t, j_t\}|=1} \left( \text{sign}(x_{n',i}, x_{n',j}; r_i^t, r_j^t) \right. \\ &\quad \left. - \text{sign}(x'_{n',i}, x'_{n',j}; r_i^t, r_j^t) \right) + L(\mathbf{a}) - L(\mathbf{a}') \end{aligned}$$

Finally, each term with  $n'$  and  $j \notin \{i_t, j_t\}$  is equal to zero, because

$$\begin{aligned} &\sum_{i \in \{i_t, j_t\}} (\text{sign}(x_{n',i}, x_{n',j}; r_i^t, r_j^t) - \text{sign}(x'_{n',i}, x'_{n',j}; r_i^t, r_j^t)) \\ &= \sum_{i \in \{i_t, j_t\}} (\text{sign}(x_{n',i}, x_{n',j}) - \text{sign}(x'_{n',i}, x'_{n',j})) \text{sign}(r_i^t, r_j^t) \\ &= (\text{sign}(x_{n',i_t}, x_{n',j}) - \text{sign}(x'_{n',i_t}, x'_{n',j})) \text{sign}(k_t, r_j^t) \\ &\quad + (\text{sign}(x_{n',j_t}, x_{n',j}) - \text{sign}(x'_{n',j_t}, x'_{n',j})) \text{sign}(k_t + 1, r_j^t) \\ &= (\text{sign}(x_{n',i_t}, x_{n',j}) - \text{sign}(x'_{n',i_t}, x'_{n',j}) + \text{sign}(x_{n',j_t}, x_{n',j}) - \text{sign}(x'_{n',j_t}, x'_{n',j})) \text{sign}(k_t, r_j^t) \\ &\quad (\text{sign}(k_t + 1, r_j^t) = \text{sign}(k_t, r_j^t)) \\ &= 0 \quad (\{x_{n',i_t}, x_{n',j_t}\} = \{x'_{n',i_t}, x'_{n',j_t}\}) \end{aligned}$$

This completes the proof of Eq. (4).

Finally, swapping the  $(i_t, j_t)$  pair only affects the score difference between  $r^t$  and  $r^{t-1}$  on pair  $(i_t, j_t)$  across the  $N-1$  peers. Hence  $U^{t-1}(\rho, \mathbf{a}) = U^t(\rho, \mathbf{a}) - 2L(\mathbf{a})$ , which proves Eq. (5).

*Proof (Proof of Lemma 3).* By the assumption of Bayesian SOC, there exist SOC conditional probabilities  $P_{n'}$  for all  $n'$  so that

$$\begin{aligned} & \Pr[\mathbf{X}_{-n} = \mathbf{X}_{-n}, \mathbf{x}_n = \mathbf{x}_n] \\ &= \sum_{\boldsymbol{\theta}} \left( \prod_{n' \neq n} P_{n'}(\mathbf{x}_{n'} | \boldsymbol{\theta}) \right) P_n(\mathbf{x}_n | \boldsymbol{\theta}) \mu(\boldsymbol{\theta}) \\ &= \frac{1}{2} \sum_{\boldsymbol{\theta}} \left( P_n(\mathbf{x}_n | \boldsymbol{\theta}) \prod_{n' \neq n} P_{n'}(\mathbf{x}_{n'} | \boldsymbol{\theta}) \right. \\ & \quad \left. + P_n(\mathbf{x}_n | \sigma(\boldsymbol{\theta})) \prod_{n' \neq n} P_{n'}(\mathbf{x}_{n'} | \sigma(\boldsymbol{\theta})) \right) \mu(\boldsymbol{\theta}) \end{aligned}$$

Recall that any  $\mathbf{X}_{-n} \in c$  can be constructed from  $\hat{\mathbf{X}}_{-n}$  and  $\check{\mathbf{X}}_{-n}$  according to the swap vector  $\mathbf{a}$ . Given  $\boldsymbol{\theta}$ , let  $p_{n'}(1) = P_{n'}(\hat{\mathbf{x}}_{n'} | \boldsymbol{\theta})$ ,  $p_{n'}(-1) = P_{n'}(\check{\mathbf{x}}_{n'} | \boldsymbol{\theta})$ , and  $\lambda_{n'} := p_{n'}(1)/p_{n'}(-1)$ . With the above notation,

$$\begin{aligned} \prod_{n' \neq n} P_{n'}(\mathbf{x}_{n'} | \boldsymbol{\theta}) &= \prod_{n' \neq n} p_{n'}(a_{n'}) \\ &= \prod_{n' \neq n} p_{n'}(-1) \cdot \prod_{n': a_{n'}=1} \lambda_{n'}, \end{aligned}$$

and because of anonymity

$$\begin{aligned} \prod_{n' \neq n} P_{n'}(\mathbf{x}_{n'} | \sigma(\boldsymbol{\theta})) &= \prod_{n' \neq n} P_{n'}(\sigma(\mathbf{x}_{n'}) | \boldsymbol{\theta}) \\ &= \prod_{n' \neq n} p_{n'}(-a_{n'}) \\ &= \prod_{n' \neq n} p_{n'}(-1) \cdot \prod_{n': a_{n'}=-1} \lambda_{n'}. \end{aligned}$$

Using the two identities above, we can write  $\Pr[\rho, L_B = d, \mathbf{x}_n] = Q(L_B = d) \Pr[\rho, \mathbf{x}_n]$  as a polynomial of  $(\lambda_{n'})_{n' \neq n}$ .

$$\begin{aligned} & \Pr[\rho, L_B = d, \mathbf{x}_n] = Q(L_B = d) \Pr[\rho, \mathbf{x}_n] \\ &= \sum_{\mathbf{X}_{-n} \in \rho, L_B(\mathbf{a}(\mathbf{X}_{-n}))=d} \Pr[\mathbf{X}_{-n} = \mathbf{X}_{-n}, \mathbf{x}_n = \mathbf{x}_n] \\ &= \frac{1}{2} \sum_{\mathbf{a}: L_B(\mathbf{a})=d} \sum_{\boldsymbol{\theta}} \left( P_n(\mathbf{x}_n | \boldsymbol{\theta}) \prod_{n' \neq n} p_{n'}(a_{n'}) \right. \\ & \quad \left. + P_n(\mathbf{x}_n | \sigma(\boldsymbol{\theta})) \prod_{n' \neq n} p_{n'}(-a_{n'}) \right) \mu(\boldsymbol{\theta}) \\ &= \frac{1}{2} \prod_{n' \neq n} p_{n'}(-1) \sum_{\boldsymbol{\theta}} \left( P_n(\mathbf{x}_n | \boldsymbol{\theta}) \sum_{\mathbf{a}: L_B(\mathbf{a})=d} \prod_{n': a_{n'}=1} \lambda_{n'} \right. \\ & \quad \left. + P_n(\mathbf{x}_n | \sigma(\boldsymbol{\theta})) \sum_{\mathbf{a}: L_B(\mathbf{a})=d} \prod_{n': a_{n'}=-1} \lambda_{n'} \right) \mu(\boldsymbol{\theta}) \end{aligned}$$



We can write the product-sum terms as elementary symmetric polynomials in the variables indexed by  $B$ , denoted by  $\lambda_B = (\lambda_{n'})_{n' \in B}$ ,

$$\begin{aligned} \sum_{\mathbf{a}: L_B(\mathbf{a})=d} \prod_{n': a_{n'}=1} \lambda_{n'} &= \left( \prod_{n' \notin B} (\lambda_{n'} + 1) \right) \sum_{S \in \binom{B}{\lfloor \frac{|B|+d}{2} \rfloor}} \prod_{n' \in S} \lambda_{n'} \\ &= \left( \prod_{n' \notin B} (\lambda_{n'} + 1) \right) \mathcal{E}_{\lfloor \frac{|B|+d}{2} \rfloor}(\lambda_B) \end{aligned}$$

and

$$\sum_{\mathbf{a}: L_B(\mathbf{a})=d} \prod_{n': a_{n'}=-1} \lambda_{n'} = \left( \prod_{n' \notin B} (\lambda_{n'} + 1) \right) \mathcal{E}_{\lfloor \frac{|B|-d}{2} \rfloor}(\lambda_B).$$

Let

$$\begin{aligned} A &= \prod_{n' \neq n} p_{n'}(-1) \cdot \left( \prod_{n' \notin B} (\lambda_{n'} + 1) \right) \\ &= \prod_{n' \in B} (p_{n'}(1) + p_{n'}(-1)) \cdot \prod_{n' \notin B, n' \neq n} p_{n'}(-1). \end{aligned}$$

Therefore,

$$\begin{aligned} &(Q(\rho, L_B = d) - Q(\rho, L_B = -d)) \Pr[\mathbf{x}_n] = \Pr[\rho, L_B = d, \mathbf{x}_n] - \Pr[\rho, L_B = -d, \mathbf{x}_n] \\ &= \frac{A}{2} \sum_{\theta} \left( P_n(\mathbf{x}_n | \theta) \mathcal{E}_{\lfloor \frac{|B|+d}{2} \rfloor}(\lambda_B) + P_n(\mathbf{x}_n | \sigma(\theta)) \mathcal{E}_{\lfloor \frac{|B|-d}{2} \rfloor}(\lambda_B) \right) \mu(\theta) \\ &\quad - \frac{A}{2} \sum_{\theta} \left( P_n(\mathbf{x}_n | \theta) \mathcal{E}_{\lfloor \frac{|B|-d}{2} \rfloor}(\lambda_B) + P_n(\mathbf{x}_n | \sigma(\theta)) \mathcal{E}_{\lfloor \frac{|B|+d}{2} \rfloor}(\lambda_B) \right) \mu(\theta) \\ &= \frac{A}{2} \sum_{\theta} \left( (P_n(\mathbf{x}_n | \theta) - P_n(\mathbf{x}_n | \sigma(\theta))) (\mathcal{E}_{\lfloor \frac{|B|+d}{2} \rfloor}(\lambda_B) - \mathcal{E}_{\lfloor \frac{|B|-d}{2} \rfloor}(\lambda_B)) \right) \mu(\theta) \end{aligned}$$

which depends on

$$(P_n(\mathbf{x}_n | \theta) - P_n(\mathbf{x}_n | \sigma(\theta))) \left( \mathcal{E}_{\lfloor \frac{|B|+d}{2} \rfloor}(\lambda_B) - \mathcal{E}_{\lfloor \frac{|B|-d}{2} \rfloor}(\lambda_B) \right). \quad (11)$$

If  $\theta$  is order consistent with  $\mathbf{x}_n$  at  $(i, j)$  (and thus consistent with  $\hat{\mathbf{x}}_{n'}$ ), by Definition 2  $P_n(\mathbf{x}_n | \theta) > P_n(\mathbf{x}_n | \sigma(\theta))$  and  $\lambda_{n'} > 1$  for all  $n'$ . Hence, by Lemma 4  $\mathcal{E}_{\lfloor \frac{|B|+d}{2} \rfloor}(\lambda_B) > \mathcal{E}_{\lfloor \frac{|B|-d}{2} \rfloor}(\lambda_B)$  and Eq. (11) is positive. Similarly, if  $\theta$  is order consistent with  $\sigma(\mathbf{x})$ , Eq. (11) is also positive, which proves Eq. (6).

Let  $\mathbf{z} = (z_1, \dots, z_m)$  be a set of  $m$  variables. The *elementary symmetric polynomial* of degree  $k$  is the sum of all distinct products of  $k$  variables in  $\mathbf{z}$ ,

$$\mathcal{E}_k(\mathbf{z}) = \sum_{1 \leq l_1 < \dots < l_k \leq m} z_{l_1} \dots z_{l_k} = \sum_{S \in \binom{[m]}{k}} \prod_{l \in S} z_l. \quad (12)$$

**Lemma 4.** If  $z_1, \dots, z_m > 1$ , and  $k \in \mathbb{Z}$  with  $k > m/2$ ,  $\mathcal{E}_k(\mathbf{z}) > \mathcal{E}_{m-k}(\mathbf{z})$ .

*Proof (Proof of Lemma 4).* The proof uses the log-concavity of elementary symmetric polynomials [10]. Given any  $i = 1, \dots, m$ , we can write

$$\mathcal{E}_k(\mathbf{z}) = z_i \mathcal{E}_{k-1}(\mathbf{z}_{-i}) + \mathcal{E}_k(\mathbf{z}_{-i}). \quad (13)$$

and  $\frac{\partial \mathcal{E}_k(\mathbf{z})}{\partial z_i} = \mathcal{E}_{k-1}(\mathbf{z}_{-i})$ . Let  $f(\mathbf{z}) = \frac{\mathcal{E}_k(\mathbf{z})}{\mathcal{E}_{m-k}(\mathbf{z})}$ .

$$\begin{aligned} \frac{\partial f}{\partial z_i} &= \frac{1}{\mathcal{E}_{m-k}(\mathbf{z})^2} (\mathcal{E}_{m-k}(\mathbf{z}) \mathcal{E}_{k-1}(\mathbf{z}_{-i}) - \mathcal{E}_{m-k-1}(\mathbf{z}_{-i}) \mathcal{E}_k(\mathbf{z})) \\ &= \frac{1}{\mathcal{E}_{m-k}(\mathbf{z})^2} (\mathcal{E}_{m-k}(\mathbf{z}_{-i}) \mathcal{E}_{k-1}(\mathbf{z}_{-i}) - \mathcal{E}_{m-k-1}(\mathbf{z}_{-i}) \mathcal{E}_k(\mathbf{z}_{-i})) \end{aligned} \quad (\text{by Eq. (13)})$$

Because  $\mathcal{E}_k$  is log concave in  $k$  and  $k > k-1 \geq m-k > m-k-1$ ,

$$\mathcal{E}_{m-k}(\mathbf{z}_{-i}) \mathcal{E}_{k-1}(\mathbf{z}_{-i}) - \mathcal{E}_{m-k-1}(\mathbf{z}_{-i}) \mathcal{E}_k(\mathbf{z}_{-i}) > 0$$

and  $\frac{\partial f}{\partial z_i} > 0$  for all  $i$ . Since  $f(\mathbf{1}) = 1$  and  $z_i > 1$  for all  $i$ ,  $f(\mathbf{z}) > 1$ , which completes the proof.

*Proof (Proof of Theorem 1).* Given  $\mathbf{x}_n = \mathbf{x}_n$  and a correction path  $r^0, \dots, r^T$  as in Definition 9, it is enough to show

$$U^t \geq_{SD} U^{t-1}, \quad \forall t = 1, \dots, T. \quad (14)$$

Then transitivity of stochastic dominance gives, for all  $w \in \mathbb{R}$ ,

$$\begin{aligned} \Pr[U_n(\text{ord}(\mathbf{x}), \Gamma(\mathbf{X}_{-n})) \geq w \mid \mathbf{x}_n] &= \Pr[U^T \geq w \mid \mathbf{x}_n] \\ &\geq \Pr[U^0 \geq w \mid \mathbf{x}_n] \\ &= \Pr[U_n(\tilde{r}, \Gamma(\mathbf{X}_{-n})) \geq w \mid \mathbf{x}_n]. \end{aligned} \quad (\text{by Eq. (14)})$$

To prove Eq. (14), by Eq. (4), define  $u(\rho, d) = U^t(\rho, \mathbf{a})$  for all  $\rho$  and  $\mathbf{a}$  with  $L(\mathbf{a}) = d$ . Then

$$\begin{aligned} &\Pr[U^t(\rho, \mathbf{a}) \geq w \mid \mathbf{x}_n] - \Pr[U^{t-1}(\rho, \mathbf{a}) \geq w \mid \mathbf{x}_n] \\ &= \sum_{\rho, d} \sum_{L(\mathbf{a})=d} Q(\rho, \mathbf{a}) (\mathbf{1}[U^t(\rho, \mathbf{a}) \geq w] - \mathbf{1}[U^{t-1}(\rho, \mathbf{a}) \geq w]) \\ &= \sum_{\rho, d} \sum_{L(\mathbf{a})=d} Q(\rho, \mathbf{a}) (\mathbf{1}[u(\rho, d) \geq w] - \mathbf{1}[u(\rho, d) - 2d \geq w]) \quad (\text{by Eq. (5)}) \\ &= \sum_{\rho, d} Q(\rho, L = d) (\mathbf{1}[u(\rho, d) \geq w] - \mathbf{1}[u(\rho, d) - 2d \geq w]). \end{aligned}$$

The terms with  $d = 0$  cancel. Pairing positive and negative margins, the difference becomes

$$\begin{aligned} &\sum_{\rho \in \mathcal{V}, d > 0} Q(\rho, L = d) (\mathbf{1}[u(\rho, d) \geq w] - \mathbf{1}[u(\rho, d) - 2d \geq w]) \\ &\quad + Q(\rho, L = -d) (\mathbf{1}[u(\rho, -d) \geq w] - \mathbf{1}[u(\rho, -d) + 2d \geq w]) \\ &= \sum_{\rho \in \mathcal{V}, d > 0} (\mathbf{1}[u(\rho, d) \geq w] - \mathbf{1}[u(\rho, d) - 2d \geq w]) (Q(\rho, L = d) - Q(\rho, L = -d)) \\ &\geq 0, \end{aligned}$$

where the equality uses Eq. (4) and the final inequality uses Lemma 3 together with  $\mathbf{1}[u(\rho, d) \geq w] \geq \mathbf{1}[u(\rho, d) - 2d \geq w]$ . This completes the proof of Eq. (14).

*Remark 1.* Though the proof of Lemma 3 is mostly algebraic, it has a probabilistic interpretation. Conditional on the magnitude, the relevant score changes are sums of independent but heterogeneous Bernoulli variables, and Bayesian SOC shifts mass toward the configurations aligned with the truthful ordering. Thus the margin under truth-telling stochastically favors positive values, yielding the one-step dominance used above.

## C Proofs and Details for Theorem 2

*Proof.* Fix an index-equivariant profile  $\alpha$ , and as in Eq. (3) let  $\tilde{D}_n = \text{comp}(\alpha_n(\mathbf{x}_n))$  denote agent  $n$ 's random reported comparison matrix. By linearity,

$$\mathbb{E}[U_n(\alpha(\mathbf{X}))] = \sum_{n' \neq n} \sum_{i < j} \mathbb{E}[\tilde{D}_n(i, j) \tilde{D}_{n'}(i, j)], \quad (15)$$

so we can focus on one peer  $n' \neq n$  and one item pair  $(i, j)$  with  $i < j$ . Let  $\sigma = \sigma_{ij}$ .

We define the *alignment factor* of strategy  $\alpha_n$  at  $(i, j)$ ,

$$\kappa_n(\mathbf{x}) := 1 - 2 \Pr[\tilde{D}_n(i, j) \neq \text{sign}(x_i, x_j) \mid \mathbf{x}_n = \mathbf{x}] \in [-1, 1].$$

We can write the expected reported comparison as  $\mathbb{E}[\tilde{D}_n(i, j) \mid \mathbf{x}_n = \mathbf{x}] = \kappa_n(\mathbf{x}) \text{sign}(x_i, x_j)$ , with  $\kappa_n \equiv 1$  for truth-telling. Moreover,  $\kappa_n(\sigma(\mathbf{x})) = \kappa_n(\mathbf{x})$  because the strategy is index-equivariant. Since the strategies' randomness is independent across agents and of all signals, conditioning on  $(\mathbf{x}_n, \mathbf{x}_{n'})$  gives

$$\mathbb{E}[\tilde{D}_n(i, j) \tilde{D}_{n'}(i, j)] = \mathbb{E}[\kappa_n(\mathbf{x}_n) \kappa_{n'}(\mathbf{x}_{n'}) \text{sign}(x_{n,i}, x_{n,j}; x_{n',i}, x_{n',j})].$$

In the first part, we prove

$$\mathbb{E}[\tilde{D}_n(i, j) \tilde{D}_{n'}(i, j)] \leq \mathbb{E}[\text{sign}(x_{n,i}, x_{n,j}; x_{n',i}, x_{n',j})], \quad (16)$$

whose right side is the same pair term under truth-telling. Partition the pairs of signal vectors into groups of four,  $\{(\mathbf{x}, \mathbf{x}'), (\sigma(\mathbf{x}), \mathbf{x}'), (\mathbf{x}, \sigma(\mathbf{x}')), (\sigma(\mathbf{x}), \sigma(\mathbf{x}'))\}$  with the representative satisfying  $\text{sign}(x_i, x_j) = \text{sign}(x'_i, x'_j) = 1$ , and let  $P(\mathbf{x}, \mathbf{x}') := \Pr[\mathbf{x}_n = \mathbf{x}, \mathbf{x}_{n'} = \mathbf{x}']$ . The contribution of one group to the left side of Eq. (16) is

$$\begin{aligned} & \kappa_n(\mathbf{x}) \kappa_{n'}(\mathbf{x}') P(\mathbf{x}, \mathbf{x}') - \kappa_n(\sigma(\mathbf{x})) \kappa_{n'}(\mathbf{x}') P(\sigma(\mathbf{x}), \mathbf{x}') \\ & - \kappa_n(\mathbf{x}) \kappa_{n'}(\sigma(\mathbf{x}')) P(\mathbf{x}, \sigma(\mathbf{x}')) + \kappa_n(\sigma(\mathbf{x})) \kappa_{n'}(\sigma(\mathbf{x}')) P(\sigma(\mathbf{x}), \sigma(\mathbf{x}')) \\ & = \kappa_n(\mathbf{x}) \kappa_{n'}(\mathbf{x}') \left( P(\mathbf{x}, \mathbf{x}') - P(\sigma(\mathbf{x}), \mathbf{x}') - P(\mathbf{x}, \sigma(\mathbf{x}')) + P(\sigma(\mathbf{x}), \sigma(\mathbf{x}')) \right) \quad (\kappa_n(\sigma(\mathbf{x})) = \kappa_n(\mathbf{x})) \\ & \leq P(\mathbf{x}, \mathbf{x}') - P(\sigma(\mathbf{x}), \mathbf{x}') - P(\mathbf{x}, \sigma(\mathbf{x}')) + P(\sigma(\mathbf{x}), \sigma(\mathbf{x}')), \end{aligned}$$

where the inequality uses that  $|\kappa_n(\mathbf{x}) \kappa_{n'}(\mathbf{x}')| \leq 1$  and the group's truthful contribution is positive,

$$\left( P(\mathbf{x}, \mathbf{x}') - P(\sigma(\mathbf{x}), \mathbf{x}') \right) + \left( P(\sigma(\mathbf{x}), \sigma(\mathbf{x}')) - P(\mathbf{x}, \sigma(\mathbf{x}')) \right) > 0,$$

by Lemma 1. Summing over groups yields Eq. (16), with equality if and only if  $\kappa_n(\mathbf{x}) \kappa_{n'}(\mathbf{x}') = 1$  on every group of positive probability. Summing Eq. (16) over item pairs and peers proves

$$\mathbb{E}[U_n(\alpha(\mathbf{X}))] \leq \mathbb{E}[U_n(\Gamma(\mathbf{X}))].$$

For the second part, when the agent  $n$  gets the highest expected score  $\mathbb{E}[U_n(\boldsymbol{\alpha}(\mathbf{X}))] = \mathbb{E}[U_n(\Gamma(\mathbf{X}))]$ , equality holds in Eq. (16) for every peer  $n'$  and every item pair  $(i, j)$ . By full support,  $\kappa_n(\mathbf{x})\kappa_{n'}(\mathbf{x}') = 1$  for almost all  $(\mathbf{x}, \mathbf{x}')$ . Hence  $|\kappa_n| = |\kappa_{n'}| = 1$ , and fixing any  $\mathbf{x}'$  shows  $\kappa_n(\mathbf{x}) = \kappa_{n'}(\mathbf{x}') = \epsilon^{ij} \in \{\pm 1\}$ , independent of the signal and common to all agents. Finally, by index-equivariance, the alignment factors agree across item pairs,  $\tilde{\mathbf{D}}_n(i, j) = \epsilon \text{sign}(x_{n,i}, x_{n,j})$  almost surely for every pair. If  $\epsilon = 1$ , every agent reports  $\Gamma_n(\mathbf{x}_n)$ . If  $\epsilon = -1$ , every agent reports the reversed ranking  $\text{rev} \circ \Gamma_n(\mathbf{x}_n)$ .

## D Subset-Assignment Proofs and Details in Section 4.1

Given an assignment  $\{S_n\}_{n=1}^N$ , we generalize the mechanism to compute the score on pairs of items shared by agents. For any two agents  $n$  and  $n'$ , let  $S_{n,n'} = S_n \cap S_{n'}$  be the set of shared items. The *generalized Kendall tau mechanism* computes the sum of consistent comparisons on shared pairs:

$$U_n(\tilde{\mathbf{r}}) = \sum_{n' \neq n} \sum_{i,j \in S_n \cap S_{n'}: i < j} \tilde{\mathbf{D}}_n(i, j) \tilde{\mathbf{D}}_{n'}(i, j),$$

which generalizes Eqs. (2) and (22) to subset-ranking and subset-top-set settings. Specifically, in the subset-ranking setting, the target function outputs the relative ordering of all items within the assigned set  $S_n$ ; in the subset-top-set setting with parameter  $K_n$ , it outputs the indices of the  $K_n$  items within  $S_n$  that have the smallest signals.

To avoid degenerate cases, we assume that every elicited pair of items is assigned to multiple agents. Formally, an assignment  $\{S_n\}_{n=1}^N$  satisfies *pairwise coverage* if for every agent  $n$  and every distinct pair of items  $i, j \in S_n$ , there exists  $n' \neq n$  so that  $i, j \in S_{n'}$ . Pairwise coverage ensures that every comparison an agent makes is checked by at least one peer. Note that when the information structure is Bayesian SOC over the full item set, for any agent  $n$ , the marginal distribution of signals restricted to pairs in  $S_n$  inherits the stochastic consistency property (Definition 2).

*Proof (Proof of Theorem 3).* The proof follows the correction path arguments established in Theorems 1 and 4. Consider a step  $t$  where agent  $n$  swaps items  $i_t, j_t \in S_n$ . The change in score depends only on the set of *active peers*

$$B_t = \{n' \neq n : i_t, j_t \in S_{n'}\}$$

who also observe both items, which is not empty by the pairwise coverage condition. Peers who do not observe the pair do not contribute to the score difference by Eq. (7).

As the Bayesian SOC property holds for the marginal distribution of signals restricted to any pair, Lemma 3 applies to the margin of the active peers  $B_t$  for rankings. For subset top-set reports, the same argument is the bucket-order proof with shape  $(K_n, |S_n| - K_n)$  restricted to active peers. Therefore each correction step satisfies  $U^t \geq_{SD} U^{t-1}$ , and composing the steps gives the result.

## E Bucket-Order Proofs and Top-Set Score Identity in Section 4.2

As in Section 3.2, we analyze the mechanism one *correcting adjacent-bucket swap* at a time for agent  $n$ 's report.

To prove stochastic dominance, we first isolate the contribution of a single peer: for an agent report  $r$  and the truthful peer report  $r_{n'} = \Gamma_{n'}(\mathbf{x}_{n'})$ , set  $u_{n'}(r) := \langle \mathbf{D}_r, \mathbf{D}_{r_{n'}} \rangle$ , so that  $U_n(r, \mathbf{X}_{-n}) = \sum_{n' \neq n} u_{n'}(r)$ .

**Definition 11 (Bucket correction path).** *Given a shape- $K^n$  report  $\tilde{r}$  and the truthful target  $r_n := \Gamma_n(\mathbf{x}_n)$ , the bucket correction path  $r^0, r^1, \dots, r^T$  is defined by  $r^0 = \tilde{r}$  and, while  $r^{t-1} \neq r_n$ , choosing adjacent buckets  $b, b+1$ , letting  $i$  be the best-signal item of  $B_{b+1}^n$  and  $j$  the worst-signal item of  $B_b^n$ , and—whenever  $x_{n,i} < x_{n,j}$ —swapping them to form  $r^t$ .*

By transitivity of  $\geq_{SD}$  it suffices to prove that each step improves the score distribution:

$$U_n(r^t, \mathbf{X}_{-n}) \geq_{SD} U_n(r^{t-1}, \mathbf{X}_{-n}) \quad \text{given } \mathbf{x}_n \text{ with } x_{n,i} < x_{n,j}. \quad (17)$$

With the magnitude and alignment defined in Definition 10, write  $U^t(\rho, \mathbf{a}) = U_n(r^t, \mathbf{X}_{-n})$  for the step- $t$  score under peer signals of magnitude  $\rho$  and alignment  $\mathbf{a}$ .

Because  $r^t = \sigma_{ij} r^{t-1}$ , bucketing commutes with relabeling, and  $\langle \cdot, \cdot \rangle$  is invariant under a common relabeling, the peer contribution satisfies

$$u_{n'}^t(\sigma_{ij} \mathbf{x}_{n'}) = u_{n'}^{t-1}(\mathbf{x}_{n'}). \quad (18)$$

The following lemma shows that we can write the difference as a linear function of the alignment vector  $\mathbf{a}$ .

**Lemma 5.** *For any  $(i_t, j_t)$ ,  $\mathbf{x}_n$ , magnitude  $\rho$  and alignment vectors  $\mathbf{a}, \mathbf{a}'$ , there is a nonnegative integer vector  $\mathbf{c}_t$  so that*

$$U^t(\rho, \mathbf{a}) - U^t(\rho, \mathbf{a}') = \frac{1}{2} \langle \mathbf{c}_t, \mathbf{a} - \mathbf{a}' \rangle \quad (19)$$

and

$$U^t(\rho, \mathbf{a}) - U^{t-1}(\rho, \mathbf{a}) = \langle \mathbf{c}_t, \mathbf{a} \rangle. \quad (20)$$

*Proof.* When  $i$  moves from bucket  $b+1$  to  $b$  and  $j$  from  $b$  to  $b+1$ , a pair's comparison can change only if it involves  $i$  or  $j$  and the other element lies in bucket  $b$  or  $b+1$ : both items cross only the boundary between buckets  $b$  and  $b+1$ , so comparisons with items in buckets  $< b$  or  $> b+1$  are unaffected. For the pair  $(i, j)$ ,  $\mathbf{D}_{r^{t-1}}(i, j) = \text{sign}(b+1, b) = -1$  and  $\mathbf{D}_{r^t}(i, j) = \text{sign}(b, b+1) = +1$ , so  $\Delta \mathbf{D}(i, j) = 2$ . For  $k \in (B_b^n \cup B_{b+1}^n) \setminus \{i, j\}$ , with  $\beta(k) \in \{b, b+1\}$ ,

$$\begin{aligned} \Delta \mathbf{D}(i, k) &= \text{sign}(b, \beta(k)) - \text{sign}(b+1, \beta(k)) = +1, \\ \Delta \mathbf{D}(j, k) &= \text{sign}(b+1, \beta(k)) - \text{sign}(b, \beta(k)) = -1, \end{aligned}$$

and  $\Delta \mathbf{D}$  vanishes on all other pairs. Thus, the difference becomes

$$\begin{aligned} u_{n'}^t - u_{n'}^{t-1} &= \langle \mathbf{D}_{r^t}, \mathbf{D}_{r_{n'}} \rangle - \langle \mathbf{D}_{r^{t-1}}, \mathbf{D}_{r_{n'}} \rangle \\ &= \langle \Delta \mathbf{D}, \mathbf{D}_{r_{n'}} \rangle \\ &= 2 \mathbf{D}_{r_{n'}}(i, j) + \sum_{k \in (B_b^n \cup B_{b+1}^n) \setminus \{i, j\}} (\mathbf{D}_{r_{n'}}(i, k) - \mathbf{D}_{r_{n'}}(j, k)). \end{aligned}$$

Now, we want to write the difference as a function of the alignment vector,  $u_{n'}^t - u_{n'}^{t-1} = c_{n'}^t a_{n'}$ . If  $i, j$  lie in one bucket of  $r_{n'}$  then  $\mathbf{D}_{r_{n'}}(i, j) = 0$  and  $\mathbf{D}_{r_{n'}}(i, k) = \mathbf{D}_{r_{n'}}(j, k)$  for all  $k$ , so  $u_{n'}^t - u_{n'}^{t-1} = 0$  and we can set

$$c_{n'}^t := 0.$$

If peer  $n'$  ranks  $i$  strictly above  $j$ , i.e.  $\beta_{r_{n'}}(i) < \beta_{r_{n'}}(j)$ ,  $\mathbf{D}_{r_{n'}}(i, j) = 1$  contributes 2. For the second term,  $\mathbf{D}_{r_{n'}}(i, k) - \mathbf{D}_{r_{n'}}(j, k)$  equals 2 when  $\beta_{r_{n'}}(i) < \beta_{r_{n'}}(k) < \beta_{r_{n'}}(j)$ , 1 when  $k$  shares a bucket with exactly one of  $i, j$ , and 0 otherwise, so summing the nonnegative terms gives  $u_{n'}^t - u_{n'}^{t-1} = c_{n'}^t$  with

$$c_{n'}^t = 2 + 2|\{k : i \prec_{n'} k \prec_{n'} j\}| + |\{k : k \sim_{n'} i \text{ xor } k \sim_{n'} j\}| \geq 2,$$

the counts over  $k \in (B_b^n \cup B_{b+1}^n) \setminus \{i, j\}$  and  $\prec_{n'}, \sim_{n'}$  denoting peer  $n'$ 's strict bucket order and same-bucket relation, depending only on the buckets that the magnitude  $\rho$  fixes. This completes the

case for  $a_{n'} = 1$ ; the case  $a_{n'} = -1$  holds by symmetry. Applying  $\sigma_{ij}$  to the peer's signal negates every sign above and hence the gain, so for every peer there is a coefficient  $(\mathbf{c}^t)_{n'} \geq 0$  such that

$$u_{n'}^t - u_{n'}^{t-1} = c_{n'}^t a_{n'}. \quad (21)$$

Summing over peers gives Eq. (20),  $U^t(\rho, \mathbf{a}) - U^{t-1}(\rho, \mathbf{a}) = \sum_{n'} (\mathbf{c}^t)_{n'} a_{n'} = \langle \mathbf{c}^t, \mathbf{a} \rangle$ . For Eq. (19), Eq. (18) gives  $u_{n'}^{t-1}(\mathbf{a}) = u_{n'}^t(\text{flip } a_{n'})$ , so  $u_{n'}^t(+1) - u_{n'}^t(-1) = (\mathbf{c}^t)_{n'}$ ; as  $u_{n'}^t$  is affine in  $a_{n'} \in \{-1, 1\}$ ,

$$U^t(\rho, \mathbf{a}) - U^t(\rho, \mathbf{a}') = \sum_{n'} (u_{n'}^t(a_{n'}) - u_{n'}^t(a'_{n'})) = \frac{1}{2} \sum_{n'} (\mathbf{c}^t)_{n'} (a_{n'} - a'_{n'}) = \frac{1}{2} \langle \mathbf{c}^t, \mathbf{a} - \mathbf{a}' \rangle.$$

For a full ranking ( $\mathbf{K} = 1^M$ ) the buckets  $B_b^n, B_{b+1}^n$  are singletons, every peer splits  $i, j$ , and  $\mathbf{c}_t = 2\mathbf{1}$ , so  $\langle \mathbf{c}_t, \mathbf{a} \rangle = 2L(\mathbf{a})$ , which is Lemma 2.

**Lemma 6.** *Given  $\boldsymbol{\theta}$  with  $\theta_i < \theta_j$ ,  $U_n(r^t, \mathbf{X}_{-n}) \geq_{SD} U_n(r^{t-1}, \mathbf{X}_{-n})$  conditional on  $\boldsymbol{\theta}$ .*

*Proof (Proof of Lemma 6).* Conditional on  $\boldsymbol{\theta}$  the peers are independent (Section 2.1). If we condition further on the magnitude  $\rho$  (Definition 10), stochastic consistency (Definition 2) gives  $p_{n'} := \Pr[a_{n'} = +1 \mid \boldsymbol{\theta}, \rho] \geq \frac{1}{2}$ , the configuration  $a_{n'} = +1$  being order consistent with  $\boldsymbol{\theta}$  at  $(i, j)$ .

Fix a peer  $n'$  and let  $u_+, u_-$  be the values of  $u_{n'}^t$  at  $a_{n'} = +1$  and  $a_{n'} = -1$ ; by Eq. (21) the gain is  $u_+ - u_- = (\mathbf{c}_t)_{n'} \geq 0$ , and by Eq. (18)  $u_{n'}^{t-1}$  equals  $u_{n'}^t$  with  $a_{n'}$  flipped, so  $u_{n'}^t$  takes the larger value  $u_+$  with probability  $p_{n'}$  whereas  $u_{n'}^{t-1}$  takes it with probability  $1 - p_{n'} \leq p_{n'}$ . As  $u_{n'}^t$  and  $u_{n'}^{t-1}$  both have support on  $\{u_+, u_-\}$  and the former has more mass on the larger,  $u_{n'}^t \geq_{SD} u_{n'}^{t-1}$  given  $(\boldsymbol{\theta}, \rho)$ .

Finally, because stochastic dominance is preserved by sums of independent random variables [24], summing over the independent coordinates gives

$$U_n(r^t) = \sum_{n'} u_{n'}^t \geq_{SD} \sum_{n'} u_{n'}^{t-1} = U_n(r^{t-1})$$

given  $(\boldsymbol{\theta}, \rho)$ ; averaging over  $\rho$  preserves the order.

*Proof (Proof of Theorem 4).* Fix an agent  $n$ , signal  $\mathbf{x}_n$ , and misreport  $\tilde{r}$ , with correction path  $r^0, \dots, r^T$  (Definition 11). By transitivity of  $\geq_{SD}$ , it suffices to prove the single-step dominance in Eq. (17).

Fix a step and a threshold  $w \in \mathbb{R}$ , and set

$$G(\boldsymbol{\theta}, w) := \Pr[U^t \geq w \mid \boldsymbol{\theta}] - \Pr[U^{t-1} \geq w \mid \boldsymbol{\theta}],$$

with  $U^s = U_n(r^s, \mathbf{X}_{-n})$ . Lemma 6 gives  $G(\boldsymbol{\theta}, w) \geq 0$  whenever  $\theta_i < \theta_j$ . Moreover, by Eq. (18), sampling  $\mathbf{x}_{n'} \sim P_{n'}(\cdot \mid \sigma_{ij}\boldsymbol{\theta})$  is the same as sampling  $\mathbf{x}_{n'} \sim P_{n'}(\cdot \mid \boldsymbol{\theta})$  and relabeling by  $\sigma_{ij}$ . Hence  $G(\sigma_{ij}\boldsymbol{\theta}, w) = -G(\boldsymbol{\theta}, w)$ .

Stochastic consistency applied to the agent's own signal, together with the exchangeability of  $\mu$ , yields the posterior tilt  $\Pr[\boldsymbol{\theta} \mid \mathbf{x}_n] \geq \Pr[\sigma_{ij}\boldsymbol{\theta} \mid \mathbf{x}_n]$  for  $\theta_i < \theta_j$ . Pairing each  $\boldsymbol{\theta}$  with  $\sigma_{ij}\boldsymbol{\theta}$ ,

$$\begin{aligned} & \Pr[U^t \geq w \mid \mathbf{x}_n] - \Pr[U^{t-1} \geq w \mid \mathbf{x}_n] \\ &= \sum_{\boldsymbol{\theta}: \theta_i < \theta_j} (\Pr[\boldsymbol{\theta} \mid \mathbf{x}_n] - \Pr[\sigma_{ij}\boldsymbol{\theta} \mid \mathbf{x}_n]) G(\boldsymbol{\theta}, w) \geq 0, \end{aligned}$$

which proves Eq. (17). Composing the  $T$  steps gives

$$U_n(r_n, \mathbf{X}_{-n}) \geq_{SD} \dots \geq_{SD} U_n(r^0, \mathbf{X}_{-n}) = U_n(\tilde{r}, \mathbf{X}_{-n}).$$

A mixed deviation is a mixture of such reports and inherits the dominance.

Theorem 4 subsumes Theorem 1 (shape  $1^M$ ) and the top-set corollary Corollary 1 (shape  $(K, M-K)$ ). The subset-assignment extension of Theorem 3 carries over unchanged—Lemma 6 runs over the active peers  $B_t = \{n' \neq n : i, j \in S_{n'}\}$ , which is nonempty under pairwise coverage—so the mechanism elicits subset bucket orders as well.

### E.1 Top-Set Score Identity

Top- $K$  reports are bucket orders of shape  $(K, M-K)$ , so their truthfulness follows directly from Theorem 4. We record the specialized score identity because it explains the top-set payment and its connection to binary Enforced Agreement.

Given two partial order relations  $R, R'$  with comparison matrices  $\mathbf{D}, \mathbf{D}' \in \{-1, 0, 1\}^{M \times M}$  in Definition 6, define the affine pairwise-loss functional

$$\ell_\tau(R, R') = \frac{1}{2} \sum_{i < j} (1 - \mathbf{D}(i, j) \mathbf{D}'(i, j)).$$

For top-set reports  $\tilde{r}_n, \tilde{r}_{n'} \subseteq [M]$  of fixed sizes, the score in Eq. (8) is equivalently

$$U_n(\tilde{\mathbf{r}}) = \sum_{n' \neq n} \sum_{i < j} \tilde{\mathbf{D}}_n(i, j) \tilde{\mathbf{D}}_{n'}(i, j), \quad (22)$$

where  $\mathbf{D}(i, j) = \mathbf{1}[i \in r, j \notin r] - \mathbf{1}[i \notin r, j \in r]$ .

**Lemma 7.** *Given sets  $r, r' \subseteq [M]$  with  $\mathbf{D} = \text{comp}(r)$  and  $\mathbf{D}' = \text{comp}(r')$ ,*

$$\sum_{i < j} \mathbf{D}(i, j) \mathbf{D}'(i, j) = M|r \cap r'| - |r||r'|.$$

*Proof.* If  $i, j$  have the same membership in  $r$  or the same membership in  $r'$ , the pair contributes zero. Agreement pairs have  $i \in r \cap r'$  and  $j \notin r \cup r'$ , contributing  $+1$ , while disagreement pairs have  $i \in r \setminus r'$  and  $j \in r' \setminus r$ , contributing  $-1$ . Hence

$$\begin{aligned} \sum_{i < j} \mathbf{D}(i, j) \mathbf{D}'(i, j) &= |r \cap r'| (M - |r| - |r'| + |r \cap r'|) \\ &\quad - (|r| - |r \cap r'|) (|r'| - |r \cap r'|) \\ &= M|r \cap r'| - |r||r'|. \end{aligned}$$

**Proposition 2 (Top-set Kendall tau and Enforced Agreement).** *For fixed set sizes  $K_n, K_{n'}$  and number of items  $M$ , the top-set Kendall tau score is affine equivalent to the Enforced Agreement (EA) score on binary inclusion reports:*

$$U_n(\tilde{\mathbf{r}}) = \frac{M}{2} \sum_{n' \neq n} S_{EA}(\tilde{y}_n, \tilde{y}_{n'}) + C,$$

where  $S_{EA}(\tilde{y}_n, \tilde{y}_{n'}) = \sum_{i=1}^M \mathbf{1}[\tilde{y}_{n,i} = \tilde{y}_{n',i}]$  and  $C$  depends on  $M, K_n, K_{n'}$  but not on the reports.

*Proof.* For binary inclusion vectors  $\tilde{y}_n, \tilde{y}_{n'}$ ,

$$S_{EA}(\tilde{y}_n, \tilde{y}_{n'}) = |\tilde{r}_n \cap \tilde{r}_{n'}| + |[M] \setminus (\tilde{r}_n \cup \tilde{r}_{n'})| = M - K_n - K_{n'} + 2|\tilde{r}_n \cap \tilde{r}_{n'}|.$$

Combining this with Lemma 7 gives

$$U_n(\tilde{\mathbf{r}}) = \sum_{n' \neq n} \left( \frac{M}{2} S_{EA}(\tilde{y}_n, \tilde{y}_{n'}) + \frac{M}{2} (K_n + K_{n'} - M) - K_n K_{n'} \right),$$

which proves the claim.



*More general partial-order targets.* The proof of Theorem 4 uses two features of bucket orders: each correcting step is a transposition  $\sigma_{ij}$  that relabels one report into another, and the resulting comparison-matrix increment is aligned with the corrected pair. More general partial-order targets can create dependencies among elicited comparisons, so extending stochastic-dominance truthfulness beyond fixed-shape bucket orders may require different arguments.

## F Additional Discussion of Assumptions and Related Work

In this section, we discuss our model assumptions and connect to the literature. First, we compare the Bayesian Stochastically Order Consistent (SOC) assumption with Strong Stochastic Transitivity (SST) and include a necessity observation for SOC. We then connect our framework to the probabilistic social choice literature through order independence, a relaxed form of independence of irrelevant alternatives.

*Additional relation to prior work* Compared to multi-task peer prediction mechanisms for categorical reports operating in discrete signal spaces, our model handles continuous underlying signals (real-valued quality scores) from which agents derive ordinal reports. This continuous setting, combined with the ordered structure of rankings, requires different analytical techniques: our proof exploits log-concavity of elementary symmetric polynomials and stochastic consistency properties rather than discrete signal correlation structures.

Our framework also bridges peer prediction and property elicitation [17]. By defining a target function that projects complex signal spaces onto structural ordinal reports (e.g., full rankings, top-sets), we may view our mechanism as a scoring rule for the ordinal shape of a belief distribution.

*SOC and Strong Stochastic Transitivity* To formalize this comparison, we say a conditional probability  $P : \Theta^M \rightarrow \Delta(\mathcal{X}^M)$  satisfies *Strong Stochastic Transitivity (SST)* if for all  $i, j, k \in [M]$  such that  $\Pr[x_i < x_j \mid \theta] > 1/2$  and  $\Pr[x_j < x_k \mid \theta] > 1/2$ , the following holds:

$$\Pr[x_i < x_k \mid \theta] \geq \max\{\Pr[x_i < x_j \mid \theta], \Pr[x_j < x_k \mid \theta]\}.$$

When SOC and SST are viewed as conditions on the same full-signal distribution, SOC is stronger than SST: SOC imposes pointwise swap inequalities on full signal vectors, while SST is a pairwise condition on comparison probabilities. However, the Bayesian SOC framework and the BPP/SST framework are not directly nested as modeling frameworks. Our framework allows heterogeneous agents and richer ordinal reports such as full rankings and top sets, while the BPP mechanism of Chen et al. [4] is designed for homogeneous pairwise-comparison assignments. Conversely, SST does not imply SOC, because a pairwise condition need not control the likelihood of each full signal vector under item swaps.

In addition to the examples in Section 2.1, the standard random binary noise model also satisfies Bayesian SOC, though it is not strict.

*Example 4 (Binary Noise Model).* Let  $\Theta = \mathcal{X} = \{0, 1\}$ . Here, the state vector  $\theta \in \{0, 1\}^M$  contains the true costs or qualities of the  $M$  items, drawn i.i.d. Each agent  $n$  observes, for each item  $i$ , a noisy signal:

$$x_{n,i} = \begin{cases} \theta_i & \text{with probability } 2/3 \\ 1 - \theta_i & \text{otherwise} \end{cases}$$

The following example illustrates only that SST does not imply SOC.

*Example 5 (RUM with Cauchy Noise).* Consider  $\boldsymbol{\theta}, \mathbf{x} \in \mathbb{R}^2$  in a RUM with Cauchy noise  $\epsilon \sim \text{Cauchy}(0, 1)$ , where the PDF is  $f(\epsilon) \propto \frac{1}{1+\epsilon^2}$ . We compare the density of the signal  $\mathbf{x}$  (which is consistent with  $\boldsymbol{\theta}$  since  $x_1 < x_2$  and  $\theta_1 < \theta_2$ ) against the swapped signal  $\mathbf{x}' = (-4, -5)$ :

$$\frac{f_{\mathbf{x}|\boldsymbol{\theta}}((-5, -4) | (0, 2))}{f_{\mathbf{x}|\boldsymbol{\theta}}((-4, -5) | (0, 2))} = \frac{(1 + (-4)^2)(1 + (-7)^2)}{(1 + (-5)^2)(1 + (-6)^2)} = \frac{(17)(50)}{(26)(37)} = \frac{850}{962} < 1.$$

Since the consistent signal is less likely than the inconsistent one, the model violates SOC. However, the model satisfies SST by [4].

*Necessity of SOC* The SOC assumption is not only a technical condition for our proof. To illustrate this, fix an item-anonymous peer-only scoring rule  $u(r, s)$  that does not depend on the underlying information structure, where  $r$  is the agent's report and  $s$  is the peer's report. For a comparison between a signal ranking  $\mathbf{x}$  and its swapped version  $\sigma(\mathbf{x})$ , call the rule *correctly oriented* if, when the peer is perfectly informative and reports the order consistent with  $\mathbf{x}$ , the rule weakly prefers reporting  $\mathbf{x}$  to reporting  $\sigma(\mathbf{x})$ . We call it *non-degenerate* if this preference is strict.

**Proposition 3 (Necessity of SOC).** *If an SOC swap inequality fails, then there is an elicitation instance in which every fixed non-degenerate correctly oriented peer-only scoring rule fails to make truthful reporting a Bayes–Nash equilibrium.*

*Proof.* Suppose SOC fails for some number of items  $M$ , state  $\boldsymbol{\theta}$ , signal ranking  $\mathbf{x}$ , and transposition  $\sigma = \sigma_{ij}$ , with  $\mathbf{x}$  order-consistent with  $\boldsymbol{\theta}$  on  $(i, j)$ :

$$P(\mathbf{x} | \boldsymbol{\theta}) < P(\sigma(\mathbf{x}) | \boldsymbol{\theta}).$$

Consider an instance whose prior is supported uniformly on the two states  $\boldsymbol{\theta}$  and  $\sigma(\boldsymbol{\theta})$ . Let the peer be perfectly informative about the state and report the order induced by that state. By item-anonymity of the signal channel,

$$P(\mathbf{x} | \sigma(\boldsymbol{\theta})) = P(\sigma(\mathbf{x}) | \boldsymbol{\theta}).$$

Thus, after observing  $\mathbf{x}$ ,

$$\Pr[\boldsymbol{\theta} | \mathbf{x}] < \Pr[\sigma(\boldsymbol{\theta}) | \mathbf{x}].$$

The agent therefore believes that a truthful peer is more likely to report the swapped order  $\sigma(\boldsymbol{\theta})$  than the order  $\boldsymbol{\theta}$ .

For brevity, identify  $\boldsymbol{\theta}$  and  $\sigma(\boldsymbol{\theta})$  with the peer reports induced by those states. Define

$$A = u(\mathbf{x}, \boldsymbol{\theta}) = u(\sigma(\mathbf{x}), \sigma(\boldsymbol{\theta})), \quad B = u(\sigma(\mathbf{x}), \boldsymbol{\theta}) = u(\mathbf{x}, \sigma(\boldsymbol{\theta})),$$

where the equalities follow from item-anonymity of the scoring rule. Correct orientation and non-degeneracy give  $A > B$ . Writing  $\pi = \Pr[\boldsymbol{\theta} | \mathbf{x}]$ , the expected-payoff gain from reporting  $\mathbf{x}$  rather than  $\sigma(\mathbf{x})$  is

$$\pi A + (1 - \pi)B - (\pi B + (1 - \pi)A) = (2\pi - 1)(A - B) < 0.$$

Thus the agent strictly prefers the swapped report to the truthful report. Truthful reporting is therefore not a Bayes–Nash equilibrium in this instance.

*SOC and Order Independence* The SOC property also connects to *order independence* in probabilistic social choice, which is a relaxed form of independence of irrelevant alternatives [29]. Specifically, a choice function specifies the probability of selecting an item from a given set. Formally, let  $F(i, A) = \Pr[\text{choose } i \mid A]$ . A choice function  $F$  is *order independent* if for all disjoint subsets  $A, B \subset [M]$  and items  $i, j \in A$  and  $k \in B$ :

$$F(i, A) > F(j, A) \iff F(k, B \cup \{i\}) < F(k, B \cup \{j\}).$$

Given a conditional probability  $P : \Theta^M \rightarrow \Delta(\mathcal{X}^M)$ , we define the induced choice function  $F_P$  such that  $F_P(i, A) = \Pr[x_i < \min_{j \in A \setminus \{i\}} x_j \mid \theta]$  which outputs the top item based on the noisy signals.

**Proposition 4.** *If  $P$  is SOC, then the induced choice function  $F_P$  is order independent.*

*Proof.* We know  $F_P(i, A) > F_P(j, A)$  if and only if  $\theta_i < \theta_j$ . Otherwise, if  $\theta_i > \theta_j$ , by Definition 2

$$\begin{aligned} F_P(i, A) - F_P(j, A) &= \Pr[x_i < \min_{k \in A \setminus \{i\}} x_k \mid \theta] - \Pr[x_j < \min_{k \in A \setminus \{j\}} x_k \mid \theta] \\ &= \sum_{\mathbf{x}: x_i < \min_{k \in A \setminus \{i\}} x_k} \Pr[\mathbf{x} \mid \theta] - \sum_{\mathbf{x}: x_i < \min_{k \in A \setminus \{i\}} x_k} \Pr[\sigma_{i,j}(\mathbf{x}) \mid \theta] \\ &= \sum_{\mathbf{x}: x_i < \min_{k \in A \setminus \{i\}} x_k} (\Pr[\mathbf{x} \mid \theta] - \Pr[\sigma_{i,j}(\mathbf{x}) \mid \theta]) < 0 \end{aligned}$$

which is a contradiction. The opposite direction is similar.

Finally, given  $\theta_i < \theta_j$ , we have

$$F(k, B \cup \{i\}) - F(k, B \cup \{j\}) = \sum_{\mathbf{x}: x_k < \min\{\min_{\ell \in B} x_\ell, x_i\}} \Pr[\mathbf{x} \mid \theta] - \sum_{\mathbf{x}: x_k < \min\{\min_{\ell \in B} x_\ell, x_j\}} \Pr[\sigma_{i,j}(\mathbf{x}) \mid \theta]$$

Terms with  $x_k < x_i$  and  $x_k < x_j$  cancel out. The remaining terms are those where  $x_k$  is smaller than one of  $\{x_i, x_j\}$  but larger than the other. Thus, the difference reduces to:

$$\sum_{\mathbf{x}: x_j \leq x_k < \min\{\min_{\ell \in B} x_\ell, x_i\}} \Pr[\mathbf{x} \mid \theta] - \sum_{\mathbf{x}: x_i \leq x_k < \min\{\min_{\ell \in B} x_\ell, x_j\}} \Pr[\mathbf{x} \mid \theta] < 0$$

by stochastic order transitivity.

## G Experimental Protocols and Additional Results

*Payment-distribution protocol* We empirically evaluate the Kendall–tau peer-prediction mechanism on both synthetic and real-world ordinal data, with the goal of testing the predicted score dominance in synthetic SOC models and comparing observed SUSHI reports with simple alternative reporting regimes. In all experiments, we fix the number of agents  $N$  and items  $M$ , compute payments using the Kendall–tau score, and compare three reporting regimes:

1. *truth-telling*, where all agents report rankings induced by their signals; for SUSHI, this means the observed human ranking;
2. *uninformed reporting*, where all agents report uniformly random permutations; and
3. *unilateral deviation*, where one agent reports a random permutation while all others report truthfully or, for SUSHI, by their observed rankings.

For each regime, we aggregate agent payments across independent trials and compare their empirical cumulative distribution functions (ECDFs).

*Datasets and ECDF setup* We first consider two standard synthetic Bayesian SOC models: a Thurstone model with i.i.d. Gaussian item costs and independent Gaussian signal noise (Example 2), where latent item costs have variance 1 and the Gaussian signal noise is governed by the variance parameter  $\eta_n$ , which we fix to  $\eta_n = 1$  for all agents, and a Mallows model centered at a latent reference ranking with dispersion parameter  $\eta_n = 0.8$  controlling the probability of Kendall-tau perturbations (Example 1). With  $M = 30$ ,  $N = 25$ , and 600 trials, the ECDFs are consistent with truthful reporting stochastically dominating both uninformed reporting and unilateral deviation in both synthetic models, in line with the theoretical guarantees.

We then repeat the same protocol on the SUSHI preference dataset, sampling  $N = 25$  human rankings over  $M = 10$  items per trial and treating the observed human ranking as the truthful/observed report. The observed-report regime is typically right-shifted relative to the other regimes, suggesting robustness to realistic, heterogeneous preference data. The corresponding ECDFs are shown in Fig. 1.

*Structured unilateral deviations* We also reuse the payment-distribution protocol but replace the single uniform-random deviation with a family of structured deviations. This tests the score against near-truthful and coordination-style reports rather than only against a uniformly random report. In each trial, we draw the truthful profile as above, or the observed-report profile for SUSHI. For a focal agent  $n$ , peers keep their truthful reports, or their observed reports for SUSHI, while agent  $n$ 's report is replaced by one of four deviations:

1. *uniform*, a uniformly random permutation, matching the baseline unilateral deviation above;
2. *partial shuffle*, the truthful or observed ranking perturbed by  $\lfloor M/2 \rfloor$  random adjacent transpositions;
3. *constant*, a fixed exogenous reference ranking reported regardless of the private signal, capturing a coordination-style report; and
4. *reverse*, the exact reversal of the truthful or observed ranking.

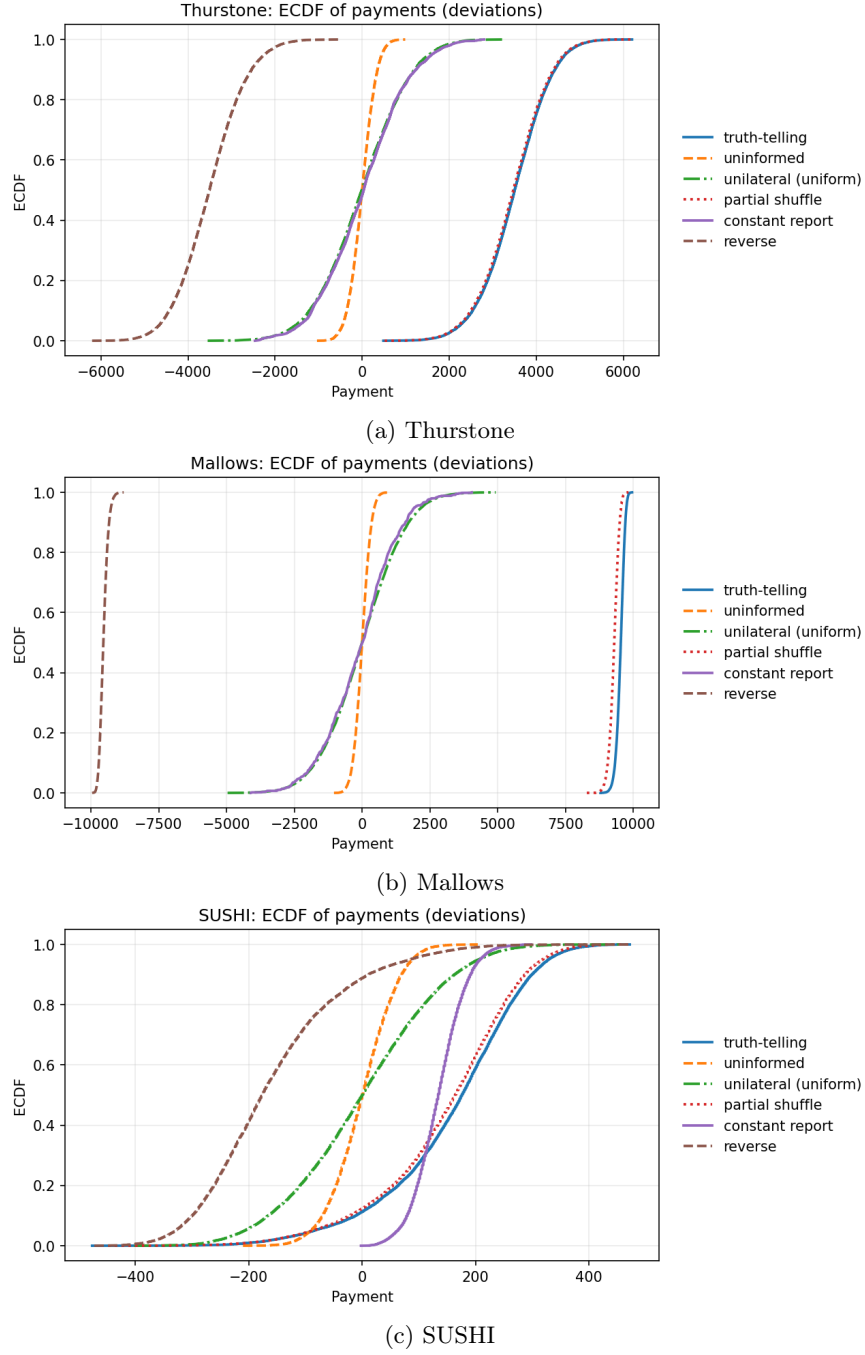
Aggregating the focal agent's Kendall tau payment over all focal agents and trials yields one empirical payment distribution per deviation. As before, lower ECDFs correspond to more favorable payment distributions.

In the synthetic Thurstone and Mallows models, truthful reporting remains right-shifted relative to these structured deviations, with the partial-shuffle deviation closest to truthful reporting and the reverse report performing worst. On SUSHI, the observed-report distribution is also generally the most favorable: although exact pointwise dominance need not hold at every threshold in this real-data stress test, the observed-report ECDF lies below or to the right of the structured-deviation curves over most of the payment range.

*Sensitivity-test details* We further examine measurement quality using a unilateral sensitivity test. Fix a reporting mechanism and designate one focal agent, while all other agents report truthfully. Let  $U_n(\eta_n)$  denote the focal agent's payment when its Thurstone signal-noise variance or Mallows dispersion parameter  $\eta_n$  is perturbed and all other agents' parameters are held fixed. We define the *sensitivity* of the mechanism as

$$\text{Sens}_n = \frac{-\frac{d}{d\eta_n} \mathbb{E}[U_n(\eta_n)]}{\sigma[U_n(\eta_n)]},$$

that is, the negative local derivative of the expected payment with respect to the focal agent's noise or dispersion parameter, evaluated at the effort baseline and normalized by the standard deviation of the payment at the baseline [30]. This yields a scale-free measure of local measurement sensitivity that is comparable across mechanisms and models.



**Fig. 3.** Structured unilateral deviations. Peers report truthfully, or by observed rankings for SUSHI, while the focal agent reports a uniform random permutation, a partially shuffled ranking, a constant ranking, or the reverse ranking.

We conduct this sensitivity test on both the Thurstone and Mallows models, using the same data-generating processes as in the payment-distribution experiments. For the Thurstone model, in each trial we first sample latent item costs  $\theta \in \mathbb{R}^M$  i.i.d. from  $\mathcal{N}(0, 1)$ . Each agent  $n$  observes these costs corrupted by independent Gaussian noise with variance parameter  $\eta_n$ , where larger  $\eta_n$  corresponds to noisier signals, and reports a full ranking that places smaller observations higher. We set the baseline noise level to  $\eta = 0.7$  for the focal agent and fix all other agents at  $\eta_{\text{others}} = 0.7$ . Here  $\eta_n$  denotes agent  $n$ 's Thurstone signal-noise variance, so larger  $\eta_n$  corresponds to lower signal quality or lower effective effort;  $\delta$  is only the finite-difference step used to estimate the local derivative. Sensitivity is estimated via symmetric finite differences with perturbation size  $\delta = 0.05$ , holding all other agents' parameters fixed. For each setting, we use 50 independent trials and 200 Monte Carlo samples to estimate both the baseline payoff standard deviation used for normalization and the finite-difference derivative, employing common random numbers to reduce variance.

For the Mallows model, in each trial we first sample a latent reference ranking uniformly at random from  $\mathfrak{S}_M$ , and each agent  $n$  independently reports a ranking drawn from a Mallows model centered at this reference, with agent-specific noise parameter  $\eta_n$  controlling the dispersion of rankings around the reference. We set the baseline dispersion to  $\eta_{\text{base}} = 0.35$  for the focal agent and fix all other agents at  $\eta_{\text{others}} = 0.35$ . An analogous symmetric finite-difference procedure is applied with perturbation size  $\delta = 0.03$ , again using 50 independent trials and 200 Monte Carlo samples for normalization and derivative estimation. In both models, we fix the number of agents to  $N = 25$ , sweep the number of items  $M$ , and evaluate how the resulting sensitivity scales with increasing ordinal complexity.

For the Thurstone sensitivity plot, we compare the Kendall–tau mechanism with six baselines. The cardinal baselines take each agent's numerical signal vector over items as input: *Enforced Agreement 1 (EA1)* discretizes each item-level signal by its sign and rewards agreement on these binary reports [32]; *mutual information (MI)* mechanisms [16] discretize numerical signals into three buckets using quantiles and score by empirical Shannon mutual information; *determinant mutual information (DMI)* uses the determinant mutual information score [13]; and *Enforced Agreement 3 (EA3)* discretizes each item-level signal by whether it is larger than the average signal for that item [32]. The ordinal baselines use only agents' reported rankings: *Enforced Agreement 2 (EA2)* coarsens each ranking into a top-half binary vector [32]; and *bonus–penalty payment (BPP)* follows the bonus–penalty adaptation of Chen et al. [4], treating sampled unordered item pairs as independent binary comparison tasks. For the Mallows sensitivity plot, we compare only with the ordinal baselines shown in the figure, EA2 and BPP.

Compared to the baselines shown in each panel, across both the Thurstone and Mallows models, the Kendall tau mechanism exhibits the largest sensitivity among the tested mechanisms as  $M$  increases (Figure 2). These results indicate that unilateral increases in noise or dispersion decrease the focal agent's expected payoff in our experimental settings, complementing the payment-distribution analysis in Section 5.